



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

Digital Forensics

Memory Forensics

A Practical Approach to Investigate Data Leakage with Volatility 2

Memory and Memory Dump

Memory Dump

- A memory dump is a process
 - takes all **information** in RAM and writes it to a storage drive in case of an application or system crash
 - known as core dump, and blue screen of death (BSOD) in Windows-based computers
- A memory dump is a binary file
- A memory dump helps software developers and system administrators to diagnose, identify and resolve the problem that led to an application or system failure.

```
A problem has been detected and windows has been shut down to prevent damage
to your computer.

The problem seems to be caused by the following file: SPCMDCON.SYS
PAGE_FAULT_IN_NONPAGED_AREA

If this is the first time you've seen this stop error screen,
restart your computer. If this screen appears again, follow
these steps:

Check to make sure any new hardware or software is properly installed.
If this is a new installation, ask your hardware or software manufacturer
for any windows updates you might need.

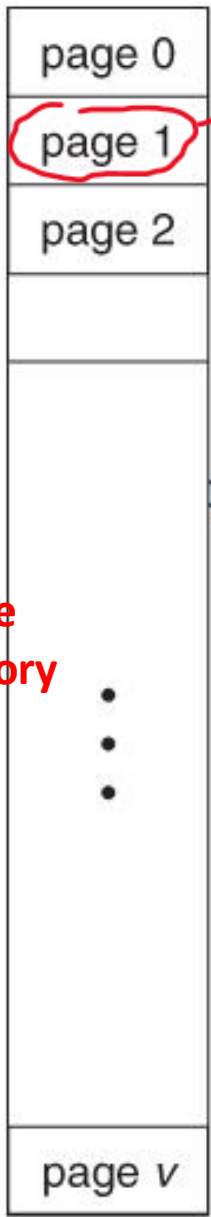
If problems continue, disable or remove any newly installed hardware
or software. Disable BIOS memory options such as caching or shadowing.
If you need to use Safe Mode to remove or disable components, restart
your computer, press F8 to select Advanced startup options, and then
select Safe Mode.

Technical information:
*** STOP: 0x00000050 (0xFD3094C2, 0x00000001, 0xFBFE7617, 0x00000000)
*** SPCMDCON.SYS - Address FBFE7617 base at FBFE5000, DateStamp 3d6dd67c
```

Memory Dump

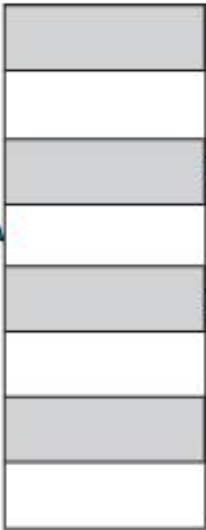
Memory, Virtual Memory, Disk

Each process can use 4GB of virtual memory



virtual memory

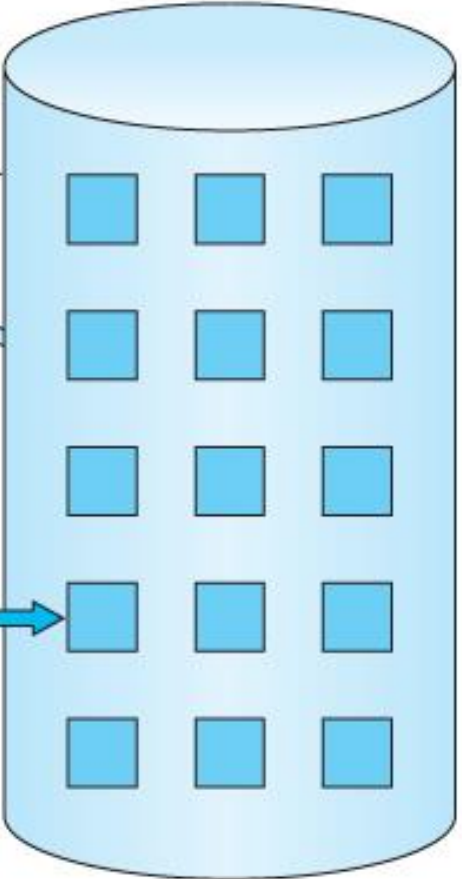
page is the smallest unit of data for memory management



memory map



physical memory



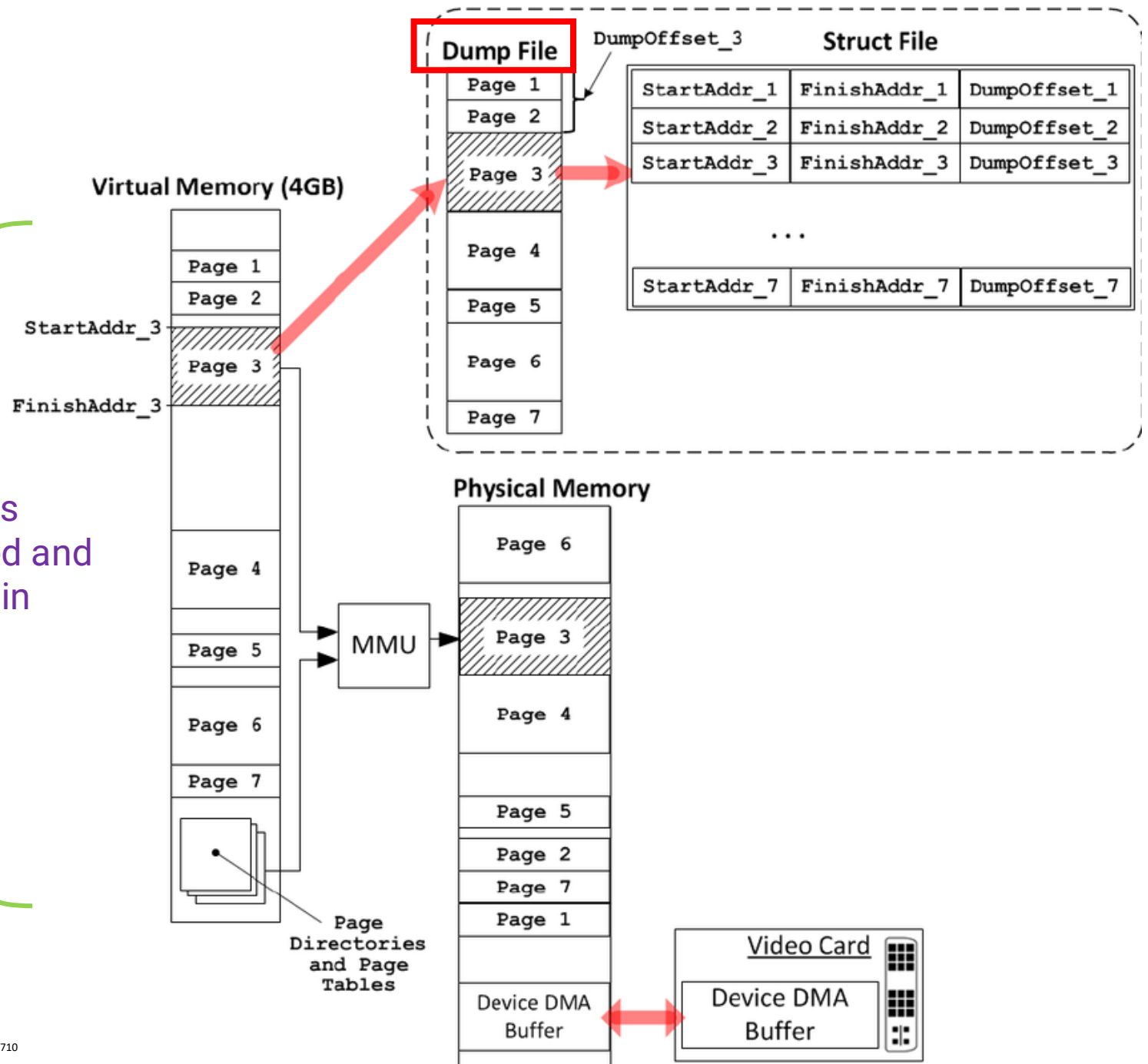
Managed by



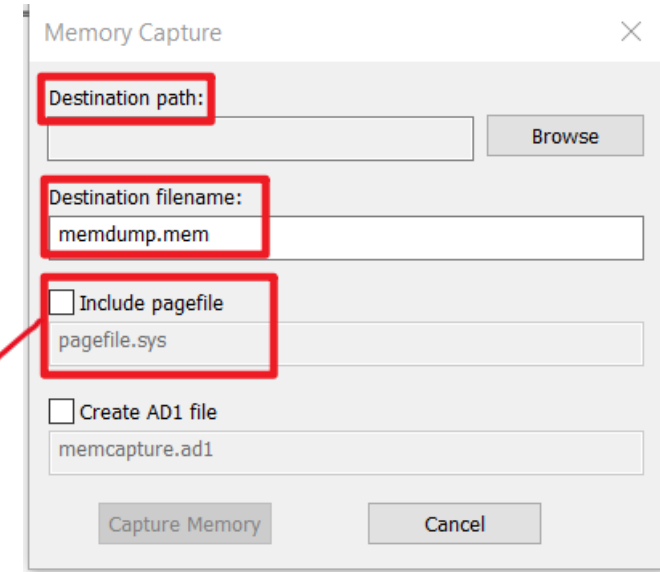
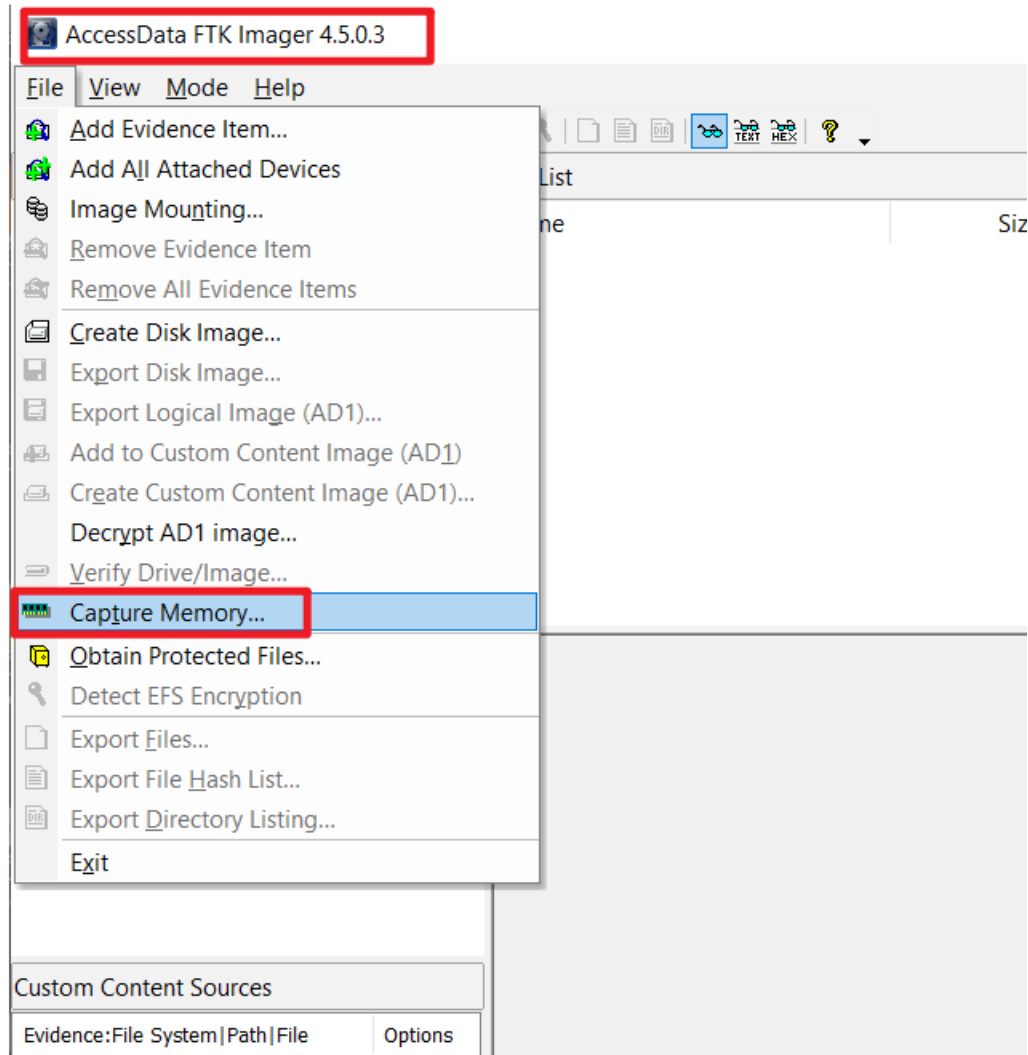
Memory dump for a process

- Kernel Processor Control Region (KPCR)
 - is a data structure used by the Windows kernel to store information about each processor
- From the KPCR structure, one then traverses intermediate structures to access interesting structures
 - such as the active process list

Each process dumped and stored in KPCR



Dump memory



- *Pagefile.sys* is a system file in Windows set aside for your computer's Random Access Memory (RAM).
- When your computer's RAM begins to run out of memory, it uses the pagefile to offload data it doesn't need, such as files and apps.

Volatility Framework - Volatile memory extraction utility framework

- For the extraction of digital artifacts from volatile memory (RAM) samples (dumps).
 - offer visibility into the runtime state of the system
 - support Win, Linux, and iOS
- Analyse raw dumps
 - **memory dumps**, crash dumps, virtual machine snapshots, VMware dumps (.vmem), Microsoft crash dumps, hibernation files, virtual box dumps
- An open collection of tools (plugins)
 - Version **2** is implemented in Python [2.6-3.0)
 - under the GNU General Public License (run, study, share, and modify)
- Provide a platform for further work in forensic research.

Pre-investigation

Volatility 2.X installation

master 11 branches 10 tags Go to file Add file <> Code

	imHlV2 Merge pull request #765 from volatilityfoundation/win...	a438e76 on Dec 11, 2020	🕒 2,826 commits
	contrib	Move unstable plugins, plugins still under development, and p...	5 years ago
	pyinstaller	Fix updated pyinstaller "empty DEST" issue	4 years ago
	resources	Move the Volatility-1.4_rc1 branch over to trunk.	12 years ago
	tools	Linux - Update check_fops for 4.20	4 years ago
	volatility	add Win10 19041 support	2 years ago
	.gitattributes	add .gitattributes	9 years ago
	.gitignore	Add in initial version of unified output plugin.	8 years ago
	AUTHORS.txt	bump credits and authors for 2.6	6 years ago
	CHANGELOG.txt	fixed to past tense	9 years ago

How to install Volatility 2?

If you have
downloaded Kali
from our GitHub,
volatility has been
pre-installed

You MUST use **vol.py**

```
(kali㉿kali)-[~]  
$ vol.py -h  
Volatility Foundation Volatility Framework 2.6.1  
Usage: Volatility - A memory forensics analysis platform.  
  
Options:  
  -h, --help                list all available options and their default values.  
                             Default values may be set in the configuration file  
                             (/etc/volatilityrc)  
  --conf-file=/home/kali/.volatilityrc  User based configuration file  
  -d, --debug                Debug volatility  
  --plugins=PLUGINS          Additional plugin directories to use (colon separated)  
  --info                      Print information about all registered objects  
  --cache-directory=/home/kali/.cache/volatility  Directory where cache files are stored  
  --cache                     Use caching  
  --tz=TZ                     Sets the (Olson) timezone for displaying timestamps  
                             using pytz (if installed) or tzset  
  -f FILENAME, --filename=FILENAME  Filename to use when opening an image  
  --profile=WinXPSP2x86          Name of the profile to load (use --info to see a list  
                             of supported profiles)  
  -l LOCATION, --location=LOCATION  A URN location from which to load an address space  
  -w, --write                  Enable write support
```

-w, --write Enable write support
--dtb=DTB DTB Address
--shift=SHIFT Mac KASLR shift address
--output=text Output in this format (support is module specific, see the Module Output Options below)
--output-file=OUTPUT_FILE Write output in this file
-v, --verbose Verbose information
--physical_shift=PHYSICAL_SHIFT Linux kernel physical shift address
--virtual_shift=VIRTUAL_SHIFT Linux kernel virtual shift address
-g KDBG, --kdbg=KDBG Specify a KDBG virtual address (Note: for 64-bit Windows 8 and above this is the address of KdCopyDataBlock)
--force Force utilization of suspect profile
--cookie=COOKIE Specify the address of nt!ObHeaderCookie (valid for Windows 10 only)
-k KPCR, --kpcr=KPCR Specify a specific KPCR address

Kernel Processor
Control Region

Supported Plugin Commands:

amcache	Print AmCache information
apihooks	Detect API hooks in process and kernel memory
atoms	Print session and window station atom tables
atomscan	Pool scanner for atom tables
auditpol	Prints out the Audit Policies from HKLM\SECURITY\Policy\PolAdtEv
bigpools	Dump the big page pools using BigPagePoolScanner
bioskbd	Reads the keyboard buffer from Real Mode memory
cachedump	Dumps cached domain hashes from memory
callbacks	Print system-wide notification routines
clipboard	Extract the contents of the windows clipboard
cmdline	Display process command-line arguments



Manual install Volatility 2? (not recommend)

```
(student@kali)-[~]
$ git clone https://github.com/volatilityfoundation/volatility.git
Cloning into 'volatility' ...
remote: Enumerating objects: 27411, done.
remote: Total 27411 (delta 0), reused 0 (delta 0), pack-reused 27411
Receiving objects: 100% (27411/27411), 21.10 MiB | 28.10 MiB/s, done.
Resolving deltas: 100% (19758/19758), done.
(student@kali)-[~]
$ ls volatility
AUTHORS.txt  CREDITS.txt  LICENSE.txt  Makefile  README.txt  pyinstaller  resources  tools  volatility
CHANGELOG.txt  LEGAL.txt  MANIFEST.in  PKG-INFO  contrib  pyinstaller.spec  setup.py  vol.py
(student@kali)-[~]
$ cd volatility
(student@kali)-[~/volatility]
$ python2 vol.py --info
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
```

for all commands starting with **volatility**, you MUST replace them with **python2 vol.py**

for example, **volatility -f memdumpWin7.mem imageinfo** MUST be changed to **python2 vol.py -f memdumpWin7.mem**

Where to download the memory image for investigations?



Download memory dump

```
student@kali:~/mem_lab$ wget https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpWin7.mem
--2020-09-19 17:18:43-- https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpwin7.mem
Resolving www.dropbox.com (www.dropbox.com)... 162.125.6.1, 2620:100:601c:1::a27d:601
Connecting to www.dropbox.com (www.dropbox.com)|162.125.6.1|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
```

Verify downloaded file

```
student@kali:~/mem_lab$ ls -l
total 1048516
-rw-r--r-- 1 student student 1073676288 Sep 19 17:18 memdumpWin7.mem
```

How to identify the image profile?

Use **vol.py** (NOT volatility) if you use the customized Kali from GitHub

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem imageinfo
Volatility Foundation Volatility Framework 2.6
INFO      : volatility.debug      : Determining profile based on KDBG search...
           Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
           AS Layer1            : IA32PagedMemoryPae (Kernel AS)
           AS Layer2            : FileAddressSpace (/home/student/mem_lab/memdumpWin7.mem)
           PAE type             : PAE
           DTB                  : 0x185000L
           KDBG                 : 0x82972c68L
           Number of Processors : 1
           Image Type (Service Pack) : 1
           KPCR for CPU 0       : 0x82973d00L
           KUSER_SHARED_DATA     : 0xffdf0000L
           Image date and time   : 2019-01-06 15:09:06 UTC+0000
           Image local date and time : 2019-01-06 07:09:06 -0800
student@kali:~/mem_lab$
```


Show all registry keys

```
(student@kalit150)-[~/mem_lab]  
$ vol.py -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey
```

Volatility Foundation volatility Framework 2.6.1

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\BCD00000000

Key name: NewStoreRoot (S)

Last updated: 2019-01-06 18:02:44 UTC+0000

Subkeys:

(S) Description

(S) Objects

Values:

Registry: \??\C:\Users\IEUser\ntuser.dat

Key name: CMI-CreateHive{6A1C4018-979D-4291-A7DC-7AED1C75B67C} (S)

Last updated: 2019-01-06 15:02:48 UTC+0000

Subkeys:

(S) AppEvents

(S) Console

(S) Control Panel

(S) Environment

Investigation



- Understand the Suspect and Accounts



- Understand the Suspect's PC



- Network Forensics



- Investigate Command History



- Investigate Suspect's USB



- Investigate Internet Explorer History



- Investigate File Explorer History



- Timeline Analysis



Understand the Suspect and Accounts

Who was using the device?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "Volatile Environment"
Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable (V) = Volatile

-----
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: Volatile Environment (V)
Last updated: 2019-01-06 15:02:48 UTC+0000

Subkeys:
(V) 1

Values:
REG_SZ LOGONSERVER : (V) \\IE8WIN7
REG_SZ USERDOMAIN : (V) IE8WIN7
REG_SZ USERNAME : (V) IEUser
REG_SZ USERPROFILE : (V) C:\Users\IEUser
REG_SZ HOMEPATH : (V) \Users\IEUser
REG_SZ HOMEDRIVE : (V) C:
REG_SZ APPDATA : (V) C:\Users\IEUser\AppData\Roaming
REG_SZ LOCALAPPDATA : (V) C:\Users\IEUser\AppData\Local
```

Physical file location = {subkeys of active user profile}

= HKEY_USERS \<SID> (linked to by HKEY_CURRENT_USER)

- SID: Security Identifier: a unique, [immutable](#) identifier of a user, user group, or other [security principal](#).
- Users can change names but not SID

Who are associated with the device?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey  
-K "Microsoft\Windows NT\CurrentVersion\ProfileList"
```

```
Volatility Foundation Volatility Framework 2.6
```

```
Legend: (S) = Stable (V) = Volatile
```

```
-----  
Registry: \SystemRoot\System32\Config\SOFTWARE
```

```
Key name: ProfileList (S)
```

```
Last updated: 2015-09-21 09:50:52 UTC+0000
```

```
Subkeys:
```

```
(S) S-1-5-18
```

```
(S) S-1-5-19
```

```
(S) S-1-5-20
```

```
(S) S-1-5-21-1716914095-909560446-1177810406-1000
```

```
(S) S-1-5-21-1716914095-909560446-1177810406-1002
```

- Sam – HKEY_LOCAL_MACHINE\SAM
- Security – HKEY_LOCAL_MACHINE\SECURITY
- **Software – HKEY_LOCAL_MACHINE\SOFTWARE**
- System – HKEY_LOCAL_MACHINE\SYSTEM
- Default – HKEY_USERS\DEFAULT

SID format details

Subkeys:

(S) S-1-5-18

(S) S-1-5-19

(S) S-1-5-20

(S) S-1-5-21-1716914095-909560446-1177810406-1000

(S) S-1-5-21-1716914095-909560446-1177810406-1002

S-1-5

S: SID

1: Version 1

5: Managed by the NT security subsystem

SID	Name	Description
S-1-5-18	Local System	A service account that is used by the operating system.
S-1-5-19	NT Authority	Local Service
S-1-5-20	NT Authority	Network Service
S-1-5-21	Administrator/Guest	A user account for the system administrator.

Who has SID= S-1-5-21-1716914095-909560446-1177810406-1002?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey  
-K "Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-1177810406-1002"
```

Volatility Foundation Volatility Framework 2.6

Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE

Key name: S-1-5-21-1716914095-909560446-1177810406-1002 (S)

Last updated: 2015-09-21 09:51:49 UTC+0000

Subkeys:

Values:

REG_EXPAND_SZ ProfileImagePath : (S) C:\Users\sshd_server

REG_DWORD Flags : (S) 0

REG_DWORD State : (S) 256

REG_BINARY Sid : (S)

0x00000000 01 05 00 00 00 00 00 05 15 00 00 00 af 07 56 66Vf

0x00000010 7e ca 36 36 e6 f5 33 46 ea 03 00 00 ~.66..3F....

REG_DWORD ProfileLoadTimeLow : (S) 0

REG_DWORD ProfileLoadTimeHigh : (S) 0

REG_DWORD RefCount : (S) 1

SSHD_Server is an account made by an SSH server., It allows remotely log in to the computer.



Who has SID= S-1-5-21-1716914095-909560446-1177810406-1000?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey  
-K "Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1716914095-909560446-11778104  
06-1000"
```

Volatility Foundation Volatility Framework 2.6

Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE

Key name: S-1-5-21-1716914095-909560446-1177810406-1000 (S)

Last updated: 2019-01-06 15:07:15 UTC+0000

Subkeys:

Values:

REG_EXPAND_SZ ProfileImagePath : (S) C:\Users\IEUser

REG_DWORD Flags : (S) 0

Who is the default logon user?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey  
-K "Microsoft\Windows NT\CurrentVersion\Winlogon"  
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile  
  
-----  
Registry: \SystemRoot\System32\Config\SOFTWARE  
Key name: Winlogon (S)  
Last updated: 2019-01-06 15:02:47 UTC+0000  
  
Subkeys:  
  (S) GPExtensions  
  (V) AutoLogonChecked  
  
Values:  
REG_SZ      ReportBootOk      : (S) 1  
REG_SZ      Shell              : (S) explorer.exe  
REG_SZ      PreCreateKnownFolders : (S) {A520A1A4-1780-4FF6-BD18-167343C5AF16}  
REG_SZ      Userinit           : (S) C:\Windows\system32\userinit.exe,  
REG_SZ      VMApplet           : (S) SystemPropertiesPerformance.exe /pagefile  
REG_SZ      WinStationsDisabled : (S) 0  
REG_DWORD   DisableCAD        : (S) 1  
REG_SZ      scremoveoption     : (S) 0  
REG_DWORD   ShutdownFlags     : (S) 2147483687  
REG_SZ      DefaultDomainName  : (S)  
REG_SZ      DefaultUserName    : (S) IEUser  
REG_SZ      AutoAdminLogon     : (S) 1
```



Default automatically log on

Hive overview

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist
Volatility Foundation Volatility Framework 2.6
Virtual      Physical    Name
-----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
```

represents pointers to other hives, and is normal

- Keys under HKLM\SYSTEM\CurrentControlSet\Control\hivelist
- Point to HKEY_LOCAL_MACHINE\SYSTEM
- Point to HKEY_LOCAL_MACHINE\HARDWARE

Each user has a profile (ntuser.dat)

It is the time to review Registry

The screenshot shows the Windows Registry Editor window. The left pane displays the tree view of the registry, with the path `Computer\HKEY_LOCAL_MACHINE\BCD00000000\Description` selected. The right pane displays a list of registry values for the selected path. Hand-drawn arrows indicate the relationship between the tree view and the labels 'Keys', 'Subkeys', and 'Values'.

Name	Type	Data
(Default)	REG_SZ	(value not set)
GuidCache	REG_BINARY	30 e0 71 da 18 b7 cf 01 06 27 00 00 14 4f 44 37 e6 e4...
KeyName	REG_SZ	BCD00000000
System	REG_DWORD	0x00000001 (1)
TreatAsSystem	REG_DWORD	0x00000001 (1)

Keys

Subkeys

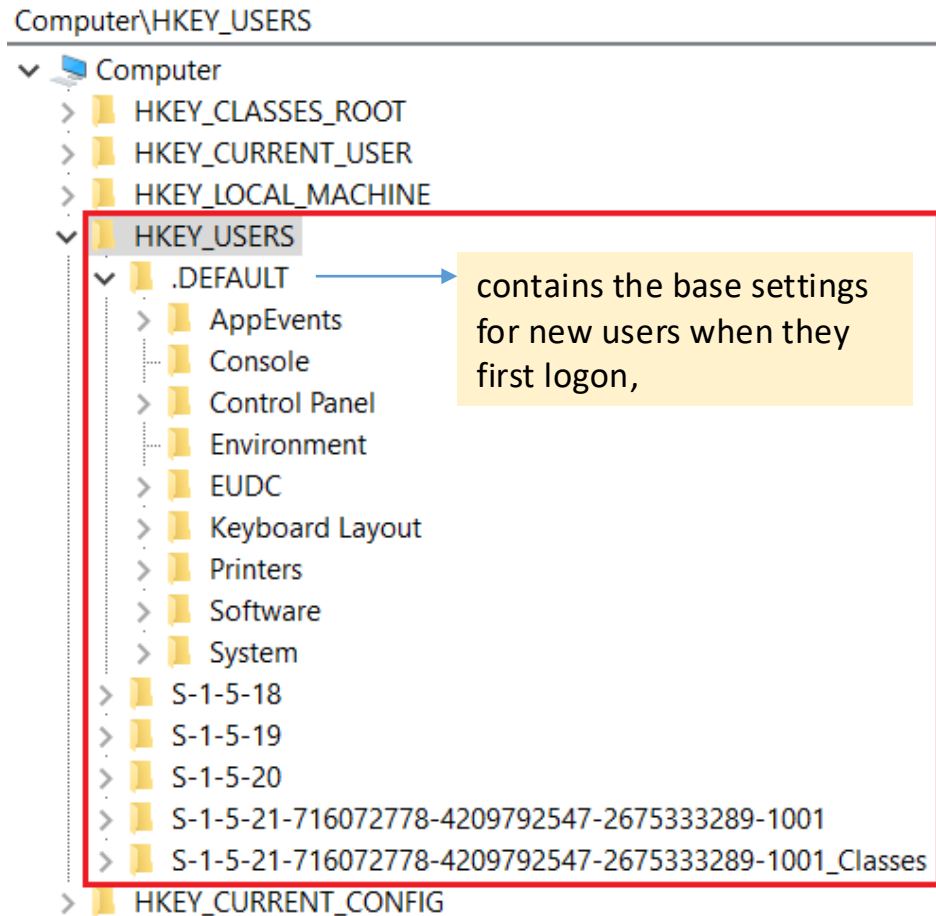
Values

The five main root keys of registry are:

- HKEY_CLASSES_ROOT (HKCR)
- HKEY_CURRENT_USER (HKCU)
- HKEY_LOCAL_MACHINE (HKLM)
- HKEY_USERS (HKU)
- HKEY_CURRENT_CONFIG (HKCC)

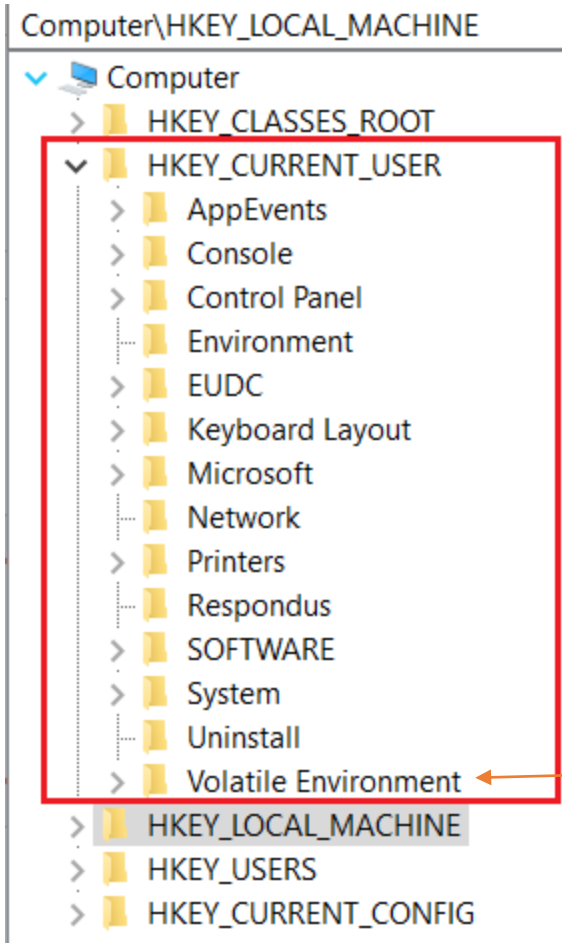
Computer\HKEY_LOCAL_MACHINE\BCD00000000\Description

HKEY_USERS (HKU)



- Contains information about all the users who logged in to the computer **at some point**
- When log on, the current logged in user profile is linked by HKCU
- Saved in each user's profile folder
 - C:\Users\IEUser\Ntuser.dat
 - C:\Users\ssh_Server\Ntuser.dat
 - C:\Users\Default User\Ntuser.dat

HKEY_CURRENT_USER (HKCU)



- Does not contain any data
 - A link to the subkey of HKEY_USERS
- Stores settings for currently logged-in use
 - Unloaded when the user logs out
 - If no profile is available, built from default user
- Control everything of the current logged user
 - Environment variables
 - Desktop settings,
 - Network connections, printers,
 - Application preferences.
 - Keyboard layout
 - Current logged user information
- Treasure for investigators

HKEY_LOCAL_MACHINE (HKLM)

- Contains computer hardware and software information
- Loaded at boot time from

The image shows a Windows Explorer window with two panes. The left pane displays the 'Computer\HKEY_LOCAL_MACHINE' tree, with 'HKEY_LOCAL_MACHINE' expanded. The right pane shows the 'Local Disk (C:) > Windows > System32 > config' directory. A red box highlights the 'HKEY_LOCAL_MACHINE' tree, and another red box highlights the 'config' directory. Arrows point from the registry keys to the corresponding files in the 'config' directory.

Computer\HKEY_LOCAL_MACHINE

- Computer
 - HKEY_CLASSES_ROOT
 - HKEY_CURRENT_USER
 - HKEY_LOCAL_MACHINE**
 - BCD00000000
 - HARDWARE
 - SAM
 - SECURITY
 - SOFTWARE
 - SYSTEM
 - HKEY_USERS
 - HKEY_CURRENT_CONFIG

Local Disk (C:) > Windows > System32 > config

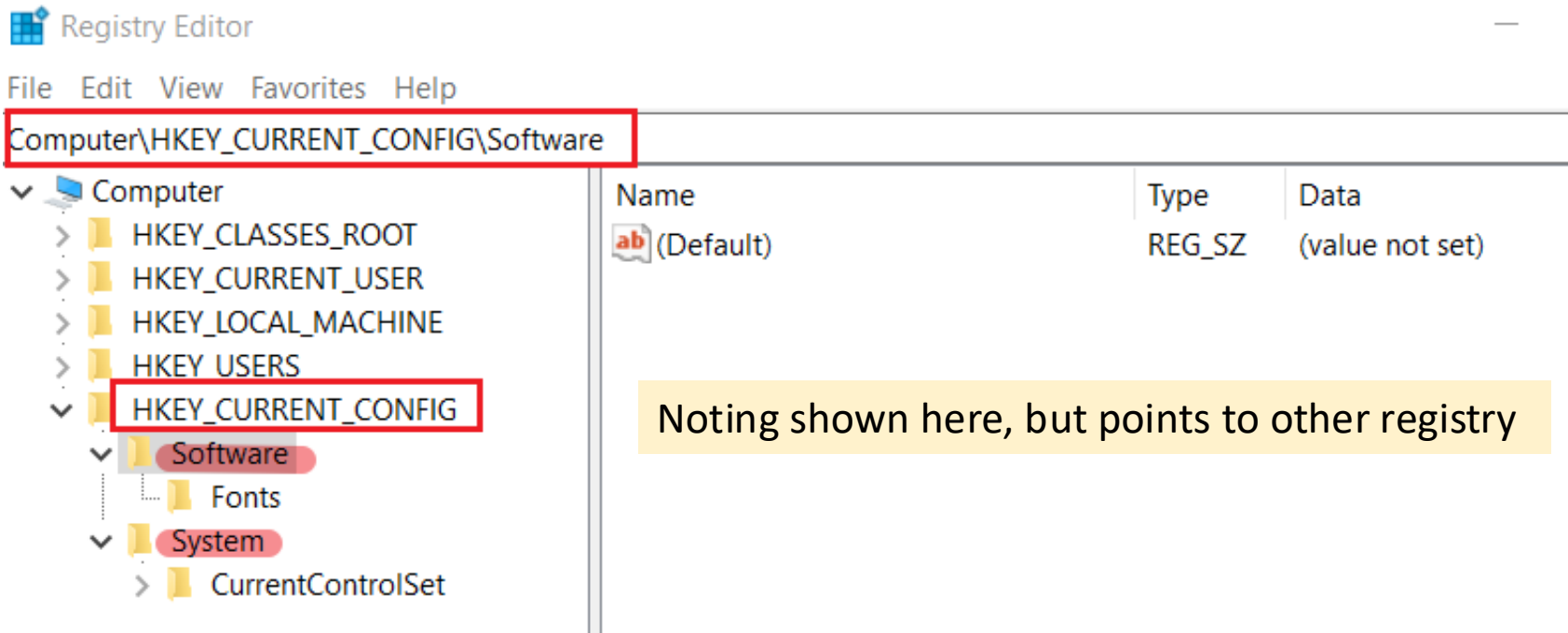
Name	Date modified	Type	Size
bbimigrate	12/19/2019 12:26 ...	File folder	
Journal	12/7/2019 4:14 AM	File folder	
RegBack	12/7/2019 4:14 AM	File folder	
systemprofile	12/7/2019 4:14 AM	File folder	
COMPONENTS	9/23/2020 9:07 AM	File	41,216 KB
DEFAULT	9/20/2020 9:12 PM	File	768 KB
DRIVERS	9/23/2020 8:15 AM	File	11,520 KB
ELAM	12/18/2019 9:33 PM	File	32 KB
SAM	9/20/2020 9:12 PM	File	128 KB
SECURITY	9/20/2020 9:12 PM	File	32 KB
SOFTWARE	9/20/2020 9:12 PM	File	113,920 KB
SYSTEM	9/20/2020 9:12 PM	File	27,392 KB
userdiff	12/19/2019 12:23 ...	File	8 KB

Loaded at boot time from

- Store Plug-and-Play devices information
- Created dynamically, not stored in a file

HKEY_CURRENT_CONFIG (HKCC)

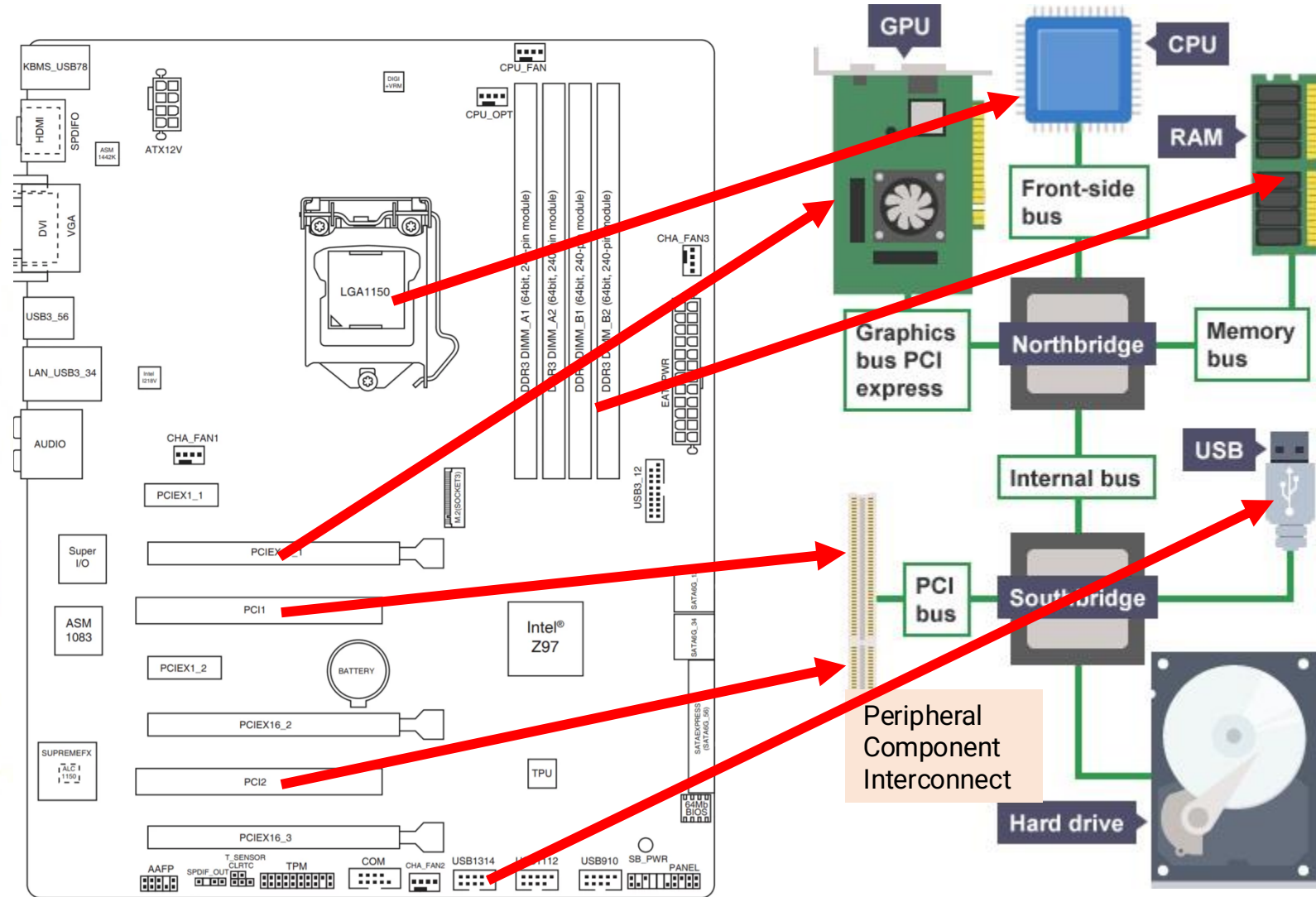
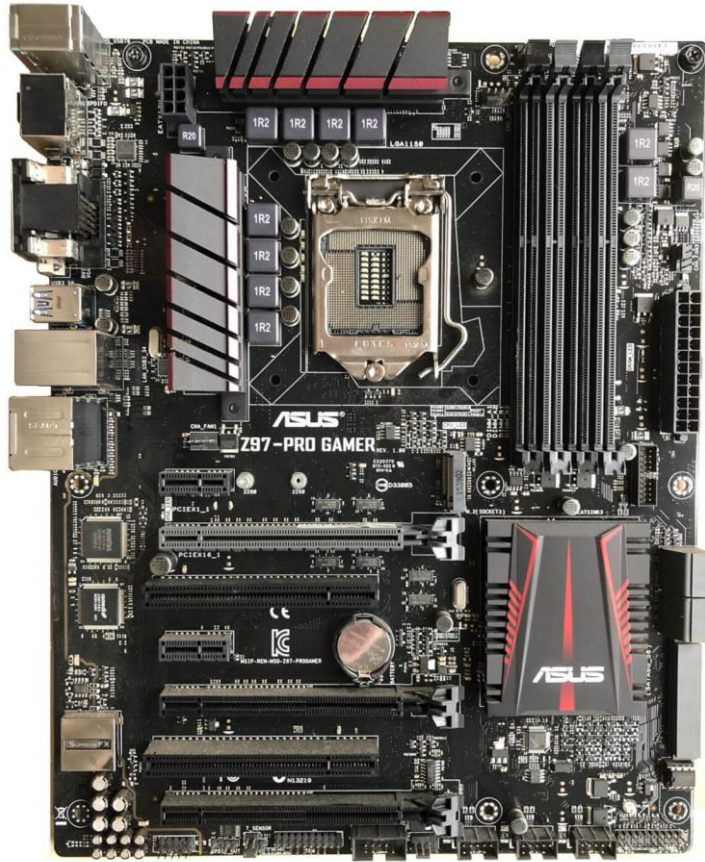
- It doesn't store any information itself but instead acts as a pointer, or a shortcut, to a registry key





Understand the Suspect's PC

Motherboard layout



Where to find CPU of the Suspect's PC?

Find hardware keys

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist
```

Volatility Foundation Volatility Framework 2.6

Virtual	Physical	Name
0x87c10370	0x27fd5370	[no name]
0x87c1c008	0x27fa3008	\REGISTRY\MACHINE\SYSTEM
0x87c459c8	0x27d8e9c8	\REGISTRY\MACHINE\HARDWARE
0x889c0430	0x1285a430	\\??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008	0x16d8c008	\\??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0	0x120049b0	\\??\C:\Users\IEUser\ntuser.dat
0x97104008	0x1c45f008	\SystemRoot\System32\Config\SECURITY
0x9711d9c8	0x1e6a09c8	\SystemRoot\System32\Config\SOFTWARE
0x981f9008	0x1b446008	\SystemRoot\System32\Config\DEFAULT
0x98209518	0x1f3a7518	\SystemRoot\System32\Config\SAM
0x982d55c0	0x1e17c5c0	\REGISTRY\MACHINE\BCD00000000
0xa620b008	0x1997b008	\\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8	0x17ad09c8	\\??\C:\Users\sshd_server\ntuser.dat
0xa6539260	0x0f690260	\\??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8	0x01c469c8	\\??\C:\System Volume Information\Syscache.hve

Point to

- HKEY_LOCAL_MACHINE
 - BCD00000000
 - HARDWARE
 - SAM
 - SECURITY
 - SOFTWARE

Find key = "DESCRIPTION\CentralProcessor\0"

Computer\HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System\CentralProcessor\0

	Name	Type	Data
Computer	(Default)	REG_SZ	(value not set)
HKEY_CLASSES_ROOT	~MHz	REG_D...	0x00000e10 (3600)
HKEY_CURRENT_USER	Component Information	REG_BI...	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
HKEY_LOCAL_MACHINE	Configuration Data	REG_FU...	ff ff ff ff ff ff 00 00 00 00 00 00 00 00
BCD00000000	FeatureSet	REG_D...	0x3d1b3fff (1025196031)
HARDWARE	Identifier	REG_SZ	Intel64 Family 6 Model 158 Stepping 9
ACPI	Platform Specific Field 1	REG_D...	0x00000002 (2)
DESCRIPTION	Previous Update Revision	REG_BI...	00 00 00 00 5e 00 00 00
System	ProcessorNameString	REG_SZ	Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz
BIOS	Update Revision	REG_BI...	00 00 00 00 b4 00 00 00
CentralProcessor	Update Status	REG_D...	0x00000000 (0)
0	VendorIdentifier	REG_SZ	GenuineIntel
1			
2			
3			
4			
5			
6			
7			
FloatingPointProcessor			
MultifunctionAdapter			
VideoAdapterBusses			
DEVICEMAP			
RESOURCEMAP			

If your system has multiple processors, they are all listed as subkeys under this key. If your system has only one processor, it is listed as processor 0.

How to find CPU of the Suspect's PC?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -o 0x87c459c8
```

```
-K "DESCRIPTION\System\CentralProcessor\0"
```

```
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile
```

Print key starts from offset
0x87c459c8. Offset can be skipped

```
-----  
Registry: \REGISTRY\MACHINE\HARDWARE
```

```
Key name: 0 (S)
```

```
Last updated: 2019-01-06 18:02:40 UTC+0000
```

```
Subkeys:
```

```
Values:
```

```
REG_BINARY    Component Information : (S)
```

```
0x00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

```
REG_SZ        Identifier           : (S) x86 Family 6 Model 158 Stepping 9
```

```
REG_FULL_RESOURCE_DESCRIPTOR Configuration Data : (S) XXXXXXXX
```

```
REG_SZ        ProcessorNameString : (S) Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz
```

```
REG_SZ        VendorIdentifier     : (S) GenuineIntel
```

```
REG_DWORD     FeatureSet           : (S) 2697805823
```

```
REG_DWORD     ~MHz                  : (S) 3600
```

```
REG_DWORD     Update Status        : (S) 2
```

```
REG_BINARY    Update Signature     : (S)
```

```
0x00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

```
REG_BINARY    Previous Update Signature : (S)
```

```
0x00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

```
REG_DWORD     Platform ID          : (S) 2
```

How to find other PC system information?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "DESCRIPTION\System"
```

```
Volatility Foundation Volatility Framework 2.6
```

```
Legend: (S) = Stable (V) = Volatile
```

```
-----  
Registry: \REGISTRY\MACHINE\HARDWARE
```

```
Key name: System (S)
```

```
Last updated: 2019-01-06 18:02:44 UTC+0000
```

```
Subkeys:
```

```
(S) CentralProcessor  
(S) FloatingPointProcessor  
(S) MultifunctionAdapter
```

Show system board, system BIOS, and video BIOS, plus subkeys for the processors and bus controllers

```
Values:
```

```
REG_BINARY    Component Information : (S)
```

```
0x00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

```
REG_SZ        Identifier : (S) AT/AT COMPATIBLE
```

```
REG_FULL_RESOURCE_DESCRIPTOR Configuration Data : (S) <REDACTED>
```

```
REG_SZ        SystemBiosDate : (S) 06/23/99
```

```
REG_MULTI_SZ  SystemBiosVersion : (S) ['VBOX - 1', '', '']
```

```
REG_DWORD     BootArchitecture : (S) 3
```

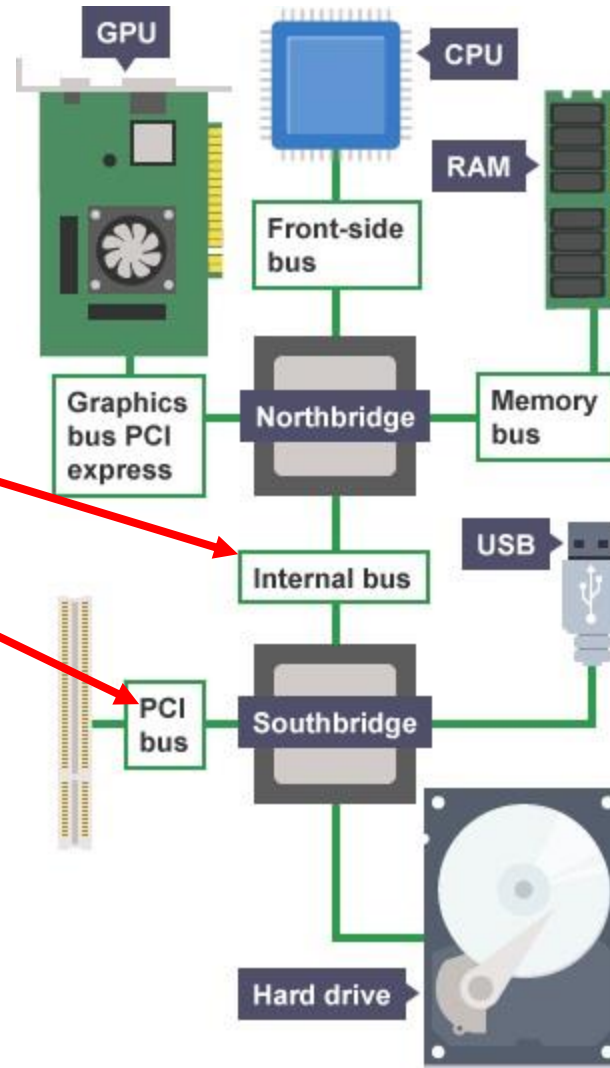
```
REG_DWORD     PreferredProfile : (S) 0
```

```
REG_DWORD     Capabilities : (S) 1345
```

```
REG_MULTI_SZ  VideoBiosVersion : (S) ['Oracle VM VirtualBox Version 6.0.0 VGA BIOS', 'Oracle VM VirtualBox Version 6.0.0 VGA BIOS', 'Oracle VM VirtualBox Version 6.0.0', '', '', '', '', '', '', '', '', '', '']
```

How PC enumerates devices on its bus (data highway)?

- A bus driver services a bridge, bus controller, or adapter
- A bus driver enumerates the devices on its bus
 - device's enumerator
- Respond to Plug and Play I/O request packets (IRPs)



Where the enumerated devices Infor saved?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 hivelist
Volatility Foundation Volatility Framework 2.6
Virtual    Physical    Name
-----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve
```

How to enumerated devices?

```
student@kali:~$ cd mem lab/  
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Enum"
```

```
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile
```

Enumerate devices

```
Registry: \REGISTRY\MACHINE\SYSTEM
```

```
Key name: Enum (S)
```

```
Last updated: 2019-01-06 15:03:07 UTC+0000
```

```
Subkeys:
```

```
(S) ACPI  
(S) DISPLAY  
(S) HTREE  
(S) IDE  
(S) LPTENUM  
(S) PCI  
(S) PCIIDE  
(S) Root  
(S) SCSI  
(S) STORAGE  
(S) SW  
(S) USBSTOR  
(S) VMBUS  
(S) WpdBusEnumRoot
```



Computer\HKEY_LOCAL_MACHINE\SYSTEM\ControlSet001\Enum

SYSTEM

ActivationBroker

ControlSet001

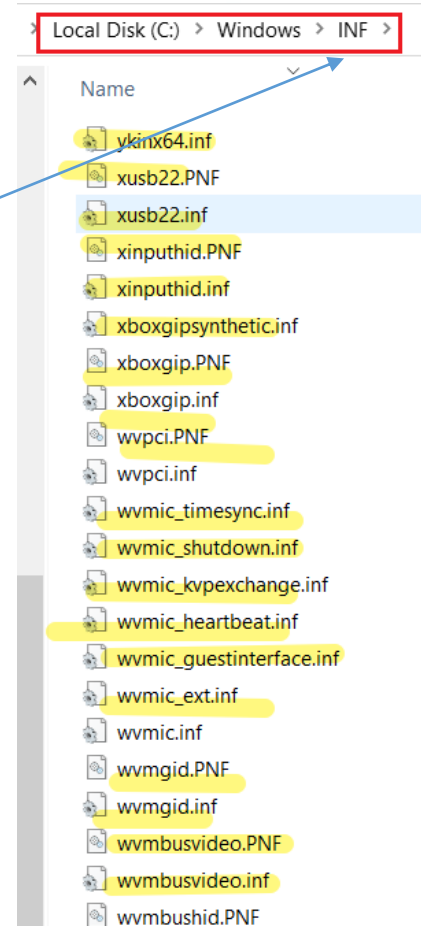
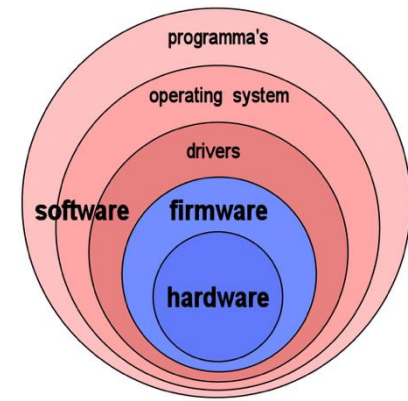
Control

Enum

- > {5d624f94-8850-40c3-a3fa-a4fd2080baf3}
- > {8e7bd593-6e6c-4c52-86a6-77175494dd8e}
- > ACPI
- > ACPI_HAL
- > BTH
- > BTHENUM
- > BTHHFENUM
- > DISPLAY
- > HDAUDIO
- > HID
- > HTREE
- > LGHUBDevice
- > PCI
- > ROOT
- > SCSI
- > STORAGE
- > SW
- > SWD
- > USB
- > USBSTOR

How Windows install device drivers?

- Almost all electronic devices contain firmware
- When plug-in, device's enumerator (e.g., PnP Bus Drivers)
 - Reports Device Identification Strings from device firmware
 - Hardware IDs - vendor-defined
 - Compatible IDs - vendor-defined
- Windows uses Hardware IDs to
 - locate the information (INF) file that best matches the device
 - if not, uses compatible IDs
- Windows uses INF file to install device drivers



How to find devices connected to PCI?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Enum\PCI"
```

```
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile
```

```
-----  
Registry: \REGISTRY\MACHINE\SYSTEM
```

```
Key name: PCI (S)
```

```
Last updated: 2019-01-06 18:02:40 UTC+0000
```

```
Subkeys:
```

```
(S) VEN_1000&DEV_0054&SUBSYS_1F091028&REV_01  
(S) VEN_1002&DEV_515E&SUBSYS_01E61028&REV_02  
(S) VEN_1022&DEV_2000&SUBSYS_20001022&REV_40  
(S) VEN_106B&DEV_003F&SUBSYS_00000000&REV_00  
(S) VEN_14E4&DEV_1659&SUBSYS_01E61028&REV_11
```

hardware ID

VEN_XXXX: vendor
DEV_XXXX: model of device

Common VEN_XXXX codes:

Intel – 8086
ATI/AMD – 1002/1022
NVIDIA – 10DE
Broadcom – 14E4
Atheros – 168C
Realtek – 10EC
Creative – 1102
Logitech – 046D

Search device: <https://devicehunt.com/>

```
(S) VEN_1000&DEV_0054&SUBSYS_1F091028&REV_01
(S) VEN_1002&DEV_515E&SUBSYS_01E61028&REV_02
(S) VEN_1022&DEV_2000&SUBSYS_20001022&REV_40
(S) VEN_106B&DEV_003F&SUBSYS_00000000&REV_00
(S) VEN_14E4&DEV_1659&SUBSYS_01E61028&REV_11
(S) VEN_15AD&DEV_0790&SUBSYS_079015AD&REV_02
(S) VEN_15AD&DEV_07A0&SUBSYS_07A015AD&REV_01
(S) VEN_8086&DEV_032C&SUBSYS_00000000&REV_09
(S) VEN_8086&DEV_100E&SUBSYS_001E8086&REV_02
(S) VEN_8086&DEV_1237&SUBSYS_00000000&REV_02
(S) VEN_8086&DEV_265C&SUBSYS_00000000&REV_00
```

Search driver based on hardware ID

VEN_8086&DEV_100E

All Maps Videos Images Shopping More

About 4,530 results (0.58 seconds)

drp.su > hwids > PCI\VEN_8086&DEV_100E

PCI\VEN_8086&DEV_100E download driver for Windows XP .

Type: PCI Vendor ID: 14E4 Device ID: 1659

Device Details

NetXtreme BCM5721 Gigabit Ethernet PCI Express

Type	Information
ID	1659

Vendor Details

Broadcom Inc. and subsidiaries

Type	Information
ID	14E4

Type: PCI Vendor ID: 8086 Device ID: 100E

Device Details

82540EM Gigabit Ethernet Controller

Type	Information
ID	100E

Vendor Details

Intel Corporation

Type	Information
ID	8086

What is the name of suspect's device?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "CurrentControlSet"
```

Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM
Key name: CurrentControlSet (V)
Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

Values:

REG_LINK SymbolicLinkValue : (V) \Registry\Machine\System\ControlSet001

Note: On a live registry, you will see "CurrentControlSet" instead of "ControlSet001". Because it is not a live, we need to check ControlSet001

link to ControlSet001

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Control\ComputerName\ComputerName"
```

Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable (V) = Volatile

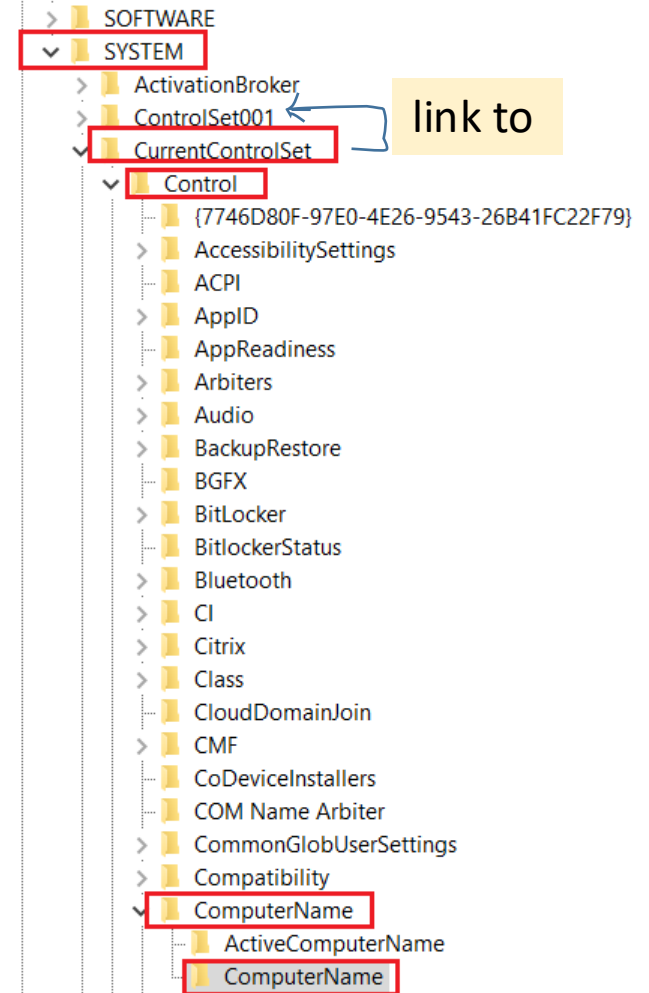
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: ComputerName (S)
Last updated: 2015-09-21 09:48:51 UTC+0000

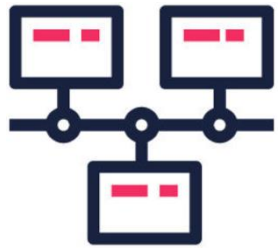
Subkeys:

Values:

REG_SZ : (S) mnmsrvc
REG_SZ ComputerName : (S) IE8WIN7

Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Cont





Network Forensics

Are there suspicious IPs (processes) connected to the PC?

Scan network objects

grep: only show TCP

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 netscan | grep TCPv4
```

Offset(P)	Proto	Local Address	Foreign Address	State	Pid	Owner	Created
0x269ff708	TCPv4	0.0.0.0:49156	0.0.0.0:0	LISTENING	440	lsass.exe	
0x3e446008	TCPv4	0.0.0.0:49154	0.0.0.0:0	LISTENING	880	svchost.exe	
0x3e446d98	TCPv4	0.0.0.0:49154	0.0.0.0:0	LISTENING	880	svchost.exe	
0x3e46e0c0	TCPv4	192.168.56.8:139	0.0.0.0:0	LISTENING	4	System	
0x3e671988	TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System	
0x3e72e530	TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	672	svchost.exe	
0x3e72ee48	TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	672	svchost.exe	
0x3e736388	TCPv4	0.0.0.0:49152	0.0.0.0:0	LISTENING	340	wininit.exe	
0x3e736d98	TCPv4	0.0.0.0:49152	0.0.0.0:0	LISTENING	340	wininit.exe	
0x3e79d528	TCPv4	0.0.0.0:49153	0.0.0.0:0	LISTENING	724	svchost.exe	
0x3e7a1618	TCPv4	0.0.0.0:49153	0.0.0.0:0	LISTENING	724	svchost.exe	
0x3e043b10	TCPv4	192.168.56.8:49177	192.168.56.5:80	CLOSED	-1		
0x3e8f0620	TCPv4	0.0.0.0:49156	0.0.0.0:0	LISTENING	440	lsass.exe	
0x3e9b3e50	TCPv4	0.0.0.0:49155	0.0.0.0:0	LISTENING	432	services.exe	
0x3ee1aa88	TCPv4	0.0.0.0:22	0.0.0.0:0	LISTENING	2016	sshd.exe	
0x3ee1bc18	TCPv4	0.0.0.0:22	0.0.0.0:0	LISTENING	2016	sshd.exe	
0x3fcba008	TCPv4	192.168.56.8:49178	192.168.56.5:80	CLOSED	-1		

Which program that was connected to this suspicious IP address?

Find out process name of 4

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 pslist -p 4
Volatility Foundation Volatility Framework 2.6
Offset(V)  Name                PID  PPID  Thds   Hnds   Sess  Wow64  Start
-----
0x8413a908 System              4    0     73    518   -----  0  2019-01-06 18:02:44 UTC+0000
```

Help command

```
student@kali:~/mem_lab$ volatility pslist -h
Volatility Foundation Volatility Framework 2.6
Usage: Volatility - A memory forensics analysis platform.

Options:
  -h, --help          list all available options and their default values.
                      Default values may be set in the configuration file
                      (/etc/volatilityrc)
```

Note: Module Output Options: dot, greptext, html, json, sqlite, text, xlsx

Which program that was connected to this suspicious IP address?

Find out process name of -1 (unidentified process) using the suspicious IP

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 yarascan -Y "192.168.56.5"  
--output-file=process_1.txt  
Volatility Foundation Volatility Framework 2.6  
Outputting to: process_1.txt  
student@kali:~/mem_lab$ ls -l  
total 1048664  
-rw-r--r-- 1 student student 1073676288 Sep 19 17:18 memdumpWin7.mem  
-rw-r--r-- 1 student student 150422 Sep 20 16:52 process_1.txt  
student@kali:~/mem_lab$
```

-Y "192.168.56.5": Define YARA pattern at IP
--output-file=any file name

YARA is a tool aimed at (but not limited to) helping malware researchers to identify and classify malware samples.

Find the application chrome.exe which PID=2388

```
student@kali:~/mem_lab$ leafpad process_1.txt
```

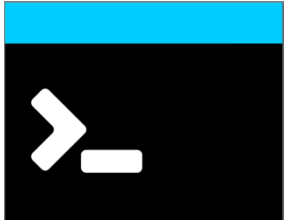
process_1.txt

File Edit Search Options Help

Owner: Process chrome.exe Pid 2388

0x010ab958	31	39	32	2e	31	36	38	2e	35	36	2e	35	2f	73	65	63	192.168.56.5/sec
0x010ab968	72	65	74	5f	66	69	6c	65	2e	64	6f	63	78	22	32	64	ret_file.docx"2d
0x010ab978	30	65	2d	35	37	65	61	34	30	62	34	37	64	39	34	63	0e-57ea40b47d94c
0x010ab988	22	46	72	69	2c	20	30	34	20	4a	61	6e	20	32	30	31	"Fri,.04.Jan.201
0x010ab998	39	20	31	36	3a	30	35	3a	31	32	20	47	4d	54	61	70	9.16:05:12.GMTap
0x010ab9a8	70	6c	69	63	61	74	69	6f	6e	2f	76	6e	64	2e	6f	70	plication/vnd.op
0x010ab9b8	65	6e	78	6d	6c	66	6f	72	6d	61	74	73	2d	6f	66	66	enxmlformats-off
0x010ab9c8	69	63	65	64	6f	63	75	6d	65	6e	74	2e	77	6f	72	64	icedocument.word
0x010ab9d8	70	6c	69	63	61	74	69	6f	6e	2f	76	6e	64	2e	6f	70	processingml.doc
0x010ab9e8	75	6a	65	64	6f	63	75	6d	65	6e	74	2e	77	6f	72	64	umentapplication
0x010ab9f8	2f	76	6e	64	2e	6f	70	65	6e	78	6d	6c	66	6f	72	6d	/vnd.openxmlform
0x010aba08	61	74	73	2d	6f	66	66	69	63	65	64	6f	63	75	6d	65	ats-officedocume
0x010aba18	6e	74	2e	77	6f	72	64	70	72	6f	63	65	73	73	69	6e	nt.wordprocessin
0x010aba28	67	6d	6c	2e	64	6f	63	75	6d	65	6e	74	83	26	03	1d	gml.document.&..
0x010aba38	00	55	61	61	06	02	02	09	08	08	0c	06	08	08	08	0d	.Uaa.....
0x010aba48	41	55	0d	0d	0d	0d	35	47	81	1b	81	1b	32	35	31	39	AU....5G....2519

The suspect used chrome to download file



Investigate Command History

cmdscan and consoles

Did the suspect use commands to copy files?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 cmdscan
Volatility Foundation Volatility Framework 2.6
*****
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2???2
Cmd #37 @ 0x3200c4: 3?4?2???2
*****
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????
```

Is F drive a USB?

When are these commands executed?

Need to find timestamps because the PID of Command Prompt

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 pslist -p 3988
Volatility Foundation Volatility Framework 2.6
Offset(V)  Name                PID  PPID  Thds  Hnds  Sess  Wow64  Start
-----
0x842ddd28 conhost.exe         3988  348   2    52    1     0  2019-01-06 15:06:11 UTC+0000
```

Command PID

An alternate method to search command history

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 consoles  
Volatility Foundation Volatility Framework 2.6  
*****
```

Process and PID found

```
*****  
ConsoleProcess: conhost.exe Pid: 3988 Process ID  
Console: 0xe181c0 CommandHistorySize: 50  
HistoryBufferCount: 2 HistoryBufferMax: 4  
OriginalTitle: %SystemRoot%\system32\cmd.exe  
Title: C:\Windows\system32\cmd.exe  
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c  
----  
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:  
CommandCount: 0 LastAdded: -1 LastDisplayed: -1  
FirstCommand: 0 CommandCountMax: 50  
ProcessHandle: 0x0  
----  
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset  
CommandCount: 3 LastAdded: 2 LastDisplayed: 2  
FirstCommand: 0 CommandCountMax: 50  
ProcessHandle: 0x5c  
Cmd #0 at 0x10dc40: ipconfig  
Cmd #1 at 0x107f90: cd Downloads  
Cmd #2 at 0x1147a0: copy secret_file.docx F:
```

Screen captured

```
Screen 0xf63b8 X:80 Y:300
Dump:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::1571:7b34:b2b1:bbe4%18
    IPv4 Address. . . . . : 192.168.56.8
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{7FD51246-C28D-46B6-AA98-EB2C150C388B}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter Teredo Tunneling Pseudo-Interface:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

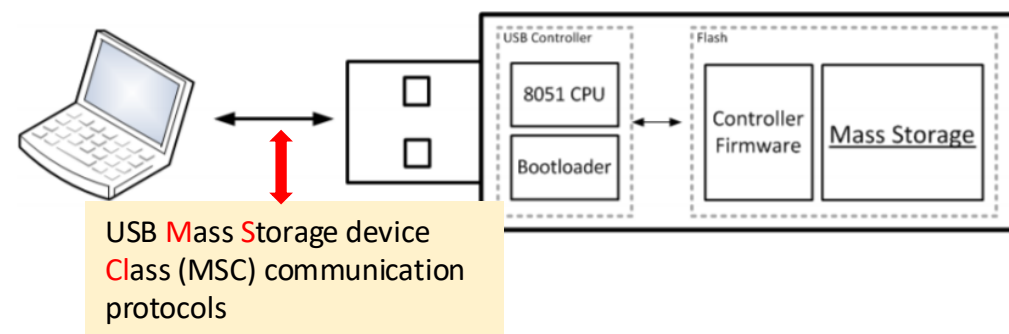
C:\Users\IEUser>cd Downloads
C:\Users\IEUser\Downloads>copy secret_file.docx F:
1 file(s) copied.

C:\Users\IEUser\Downloads>
student@kali:~/mem_lab$
```



Investigate Suspect's USB

How a PC recognizes a USB?



- When a USB is plugged in, the Plug and Play (PnP) Manager finds device ID (Hardware IDs or Compatible IDs)
 - Receives the event => enumerates devices from firmware=> device ID
- OS locates the appropriate driver for the device
 - Queries the device descriptor (*.inf*) uses Hardware IDs or Compatible IDs
- OS loads that driver use *.inf*
- Once the device has been identified
 - Multiple registry keys will be generated

Generated registry keys can be used as evidence chain

- Show a USB was attached to a PC
- A USB has an unique **Instance ID** of each device
- The USB has been assigned a drive letter (D, E, F..) of the PC
- The drive letter is associated with a suspect's account
- Timestamps need to be verified
- Check setupapi.log (if available)

Show a USB was attached to a PC

- USB Device interface key is add under
 - HKEY_LOCAL_MACHINE\SYSTEM\ControlSet001\Control\DeviceClasses
- USB Device interface key have a link to
 - USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00

What is a Device interface GUID?

- Each device has an interface with a Globally Unique Identifier (GUID)
 - To communicate with PC
 - Organized under *DeviceClasses*
- A device interface subkey is created when a plug and play device driver is successfully loaded.
 - Disk interface subkey: GUID_DEVINTERFACE_DISK, or DiskClassGUID
 - Value=53F56307-B6BF-11D0-94F2-00A0C91EFB8B
 - The subkey will be removed if the device is removed

Attribute	Setting
Identifier	a hard disk storage (e.g., USB) device interface GUID_DEVINTERFACE_DISK
Class GUID	{53F56307-B6BF-11D0-94F2-00A0C91EFB8B}

GUID_DEVICE_BATTERY
GUID_DEVICE_LID
GUID_DEVICE_MEMORY
GUID_DEVICE_MESSAGE_INDICATOR
GUID_DEVICE_PROCESSOR
GUID_DEVICE_SYS_BUTTON
GUID_DEVICE_THERMAL_ZONE
GUID_DEVINTERFACE_BRIGHTNESS
GUID_DEVINTERFACE_CDCHANGER
GUID_DEVINTERFACE_CDROM
GUID_DEVINTERFACE_COMPORT
GUID_DEVINTERFACE_DISK
GUID_DEVINTERFACE_DISPLAY_ADAPTER
GUID_DEVINTERFACE_FLOPPY
GUID_DEVINTERFACE_HID
GUID_DEVINTERFACE_I2C
GUID_DEVINTERFACE_IMAGE
GUID_DEVINTERFACE_KEYBOARD
GUID_DEVINTERFACE_MEDIUMCHANGER
GUID_DEVINTERFACE_MODEM

List all device interfaces

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Control\DeviceClasses"
```

```
Volatility Foundation Volatility Framework 2.6
```

```
Legend: (S) = Stable (V) = Volatile
```

```
-----
```

```
Registry: \REGISTRY\MACHINE\SYSTEM
```

```
Key name: DeviceClasses (S)
```

```
Last updated: 2019-01-06 15:03:08 UTC+0000
```

```
Subkeys:
```

```
(S) {0e0b6031-5213-4934-818b-38d90ced39db}
```

```
(S) {10497b1b-ba51-44e5-8318-a65c837b6661}
```

```
(S) {2accfe60-c130-11d2-b082-00a0c91efb8b}
```

```
(S) {2e34d650-5819-42ca-84ae-d30803bae505}
```

```
(S) {32412632-86cb-44a2-9b5c-50d1417354f5}
```

```
(S) {3375baf4-9e15-4b30-b765-67acb10d607b}
```

```
(S) {34d14be3-dee4-41c8-9ae7-6b174977c192}
```

```
(S) {35fa2e29-ea23-4236-96ae-3a6ebacba440}
```

```
(S) {3abf6f2d-71c4-462a-8a92-1e6861e6af27}
```

```
(S) {4116f60b-25b3-4662-b732-99a6111edc0b}
```

```
(S) {4afa3d53-74a7-11d0-be5e-00a0c9062857}
```

```
(S) {4d1e55b2-f16f-11cf-88cb-001111000030}
```

```
(S) {4d36e978-e325-11ce-bfc1-08002be10318}
```

```
(S) {53f56307-b6bf-11d0-94f2-00a0c91efb8b}
```

```
(S) {53f56308-b6bf-11d0-94f2-00a0c91efb8b}
```

```
(S) {53f5630d-b6bf-11d0-94f2-00a0c91efb8b}
```

Disk interfaces

List Disk Interfaces

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c  
91efb8b}"
```

Volatility Foundation Volatility Framework 2.6

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: {53f56307-b6bf-11d0-94f2-00a0c91efb8b} (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

(S) ###?#IDE#DiskVBOX_HARDDISK_____1.0_____#58106af171808
1.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

(S) ###?#IDE#DiskVirtual_HD_____1.1.0_____#5835dc7040808
0.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

(S) ###?#IDE#DiskVMware_Virtual_IDE_Hard_Drive_____00000001#582eba498080.
0.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

(S) ###?#SCSI#Disk&Ven_Dell&Prod_VIRTUAL_DISK#6817b13437808000000#{53f56307-b6b
f-11d0-94f2-00a0c91efb8b}

(S) ###?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#681bec0f48808_80#{53f5630
7-b6bf-11d0-94f2-00a0c91efb8b}

#USBSTOR

#Disk

&Ven_[VendorName]

&Prod_[ProductName]

&Rev_[RevisionID]

#[UID/SerialNo.]

#[DeviceInterfaceClassID].

Other Useful **disk** interfaces (1)

- **Tableau connected via Firewire:**

##?#SBP2#Tableau&Forensic_SATA_Bridge&LUN0&REV15#000ecc01000f4063#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

- **eSATA attached SSD:**

##?#IDE#DiskPatriot_Torqx_2_32GB_SSD_____S5FAM014#4&35e86db3&0&0.2.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

- **F-Response iSCSI disk:**

##?#SCSI#Disk&Ven_FRES&Prod_FRES#1&1c121344&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

- **External non generic USB Disk:**

\##?#USBSTOR#Disk&Ven_FUJITSU&Prod_MHT2060AH&Rev_#43527242060A&0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

- **External generic USB Disk:**

##?#USBSTOR#Disk&Ven_&Prod_&Rev_PMAP#07032AE8AFBE2481&0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

Other Useful **disk** interfaces (2)

- **USB attached android phone:**

##?#USBSTOR#Disk&Ven_Android&Prod___UMS_Composite&Rev___00#8&d3cc3f0&0&304D1905736FB98E&0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

- **Volume GUIDs:** SYSTEM\\Control\\DeviceClasses\\{53F5630D-B6BF-11D0-94F2-00A0C91EFB8B}

- **MagicISO mounted image:** ##?#SCSI#CdRom&Ven_MagicISO&Prod_Virtual_DVD-ROM&Rev_1.0A#1&2afd7d61&0&0000#{53f5630d-b6bf-11d0-94f2-00a0c91efb8b}

- **Volume of the generic USB storage device:**

##?#STORAGE#VOLUME#_??_USBSTOR#DISK&VEN_&PROD_&REV_PMAP#07032AE8AFBE2481&0#{53F56307-B6BF-11D0-94F2-00A0C91EFB8B}#{53f5630d-b6bf-11d0-94f2-00a0c91efb8b}

Access DeviceClass\Disk Interface\USB Interface

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c  
91efb8b}\##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f5  
6307-b6bf-11d0-94f2-00a0c91efb8b}"
```

Volatility Foundation Volatility Framework 2.6

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: ##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f
56307-b6bf-11d0-94f2-00a0c91efb8b} (S)

Last updated: 2019-01-06 15:02:54 UTC+0000 *timestamp*

Subkeys:

(S) #

(V) Control

DeviceInstance is saved in anther registry

Values:

```
REG_SZ      DeviceInstance : (S) USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00  
\6&1bec0f48&0&_&0
```

Access DeviceClass\Disk Interface\USB Interface\# SymbolicLink

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91  
efb8b}\##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f5630  
7-b6bf-11d0-94f2-00a0c91efb8b}\#"
```

Volatility Foundation Volatility Framework 2.6

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: # (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

USB plug-in timestamp

Subkeys:

(V) Control

It is a symbolic link, we can check and verify the subkey

Values:

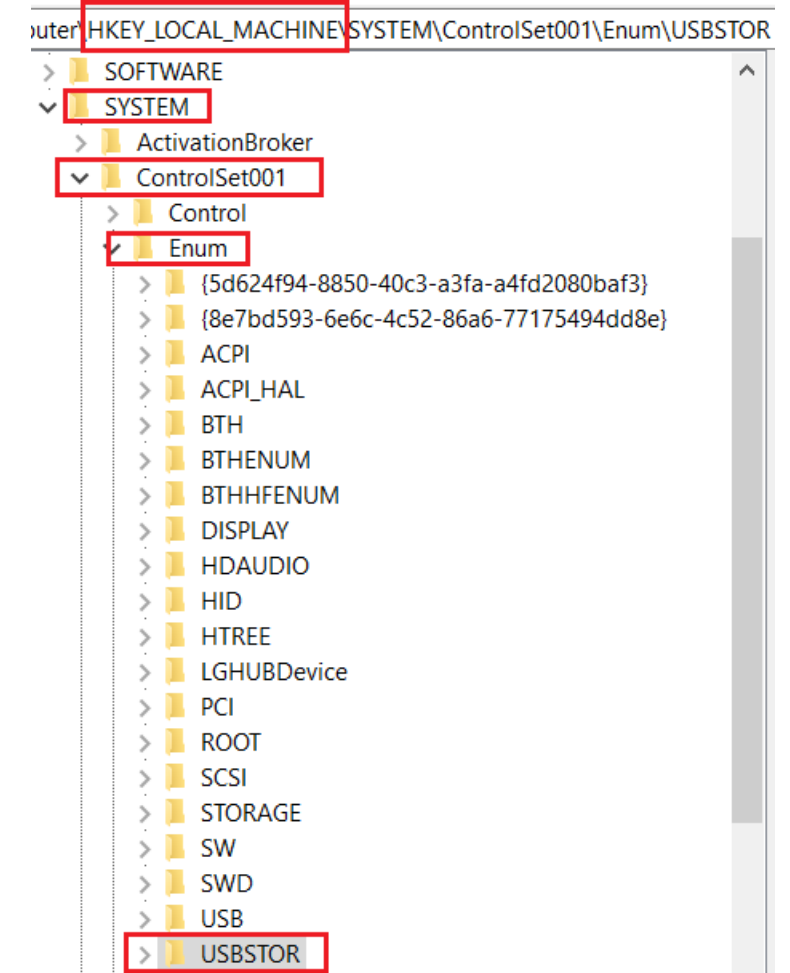
```
REG_SZ      SymbolicLink      : (S) \\?\USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5  
.00#6&1bec0f48&0&_&0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
```

Verify USBTOR subkey and timestamp

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Enum\USBSTOR"  
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile  
  
-----  
Registry: \REGISTRY\MACHINE\SYSTEM  
Key name: USBSTOR (S)  
Last updated: 2019-01-06 15:02:54 UTC+0000  
  
Subkeys:  
(S) Disk&Ven_General&Prod_UDisk&Rev_5.00
```

USB Storage

Same timestamp



Ven_ [vendor name]

Prod_ [product name assigned by vendor]

Rev_ [revision ID]

Can we find Vendor ID and Product ID?

How to find Vendor ID and Product ID?

- When a USB device is connected to a Windows PC, Windows will remember its vendor and class IDs to reference the next time it is plugged in. This information is stored in:
 - HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Enum\USB
 - E.g., Vid_**0c55**&Pid_**5416**
- Can search device information
 - <https://www.the-sz.com/products/usbid/>
- Note: Registry dumped from memory doesn't have VID and PID
 - Normal registry will contain these information

What is the Instance ID of USB?

- A device identification string that distinguishes a device from other devices of the same type on a computer.

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00"  
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile  
  
-----  
Registry: \REGISTRY\MACHINE\SYSTEM  
Key name: Disk&Ven_General&Prod_UDisk&Rev_5.00 (S)  
Last updated: 2019-01-06 15:02:54 UTC+0000  
  
Subkeys:  
(S) 6&1bec0f48&0&_&0
```

the second character is &, means

- the USB doesn't provide SN
- UID is created by PnP, is only unique system wide

What is the friendly name of the USB?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey  
-K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_50"  
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile  
  
-----  
Registry: \REGISTRY\MACHINE\SYSTEM  
Key name: 6&1bec0f48&0&_50 (S)  
Last updated: 2019-01-06 15:02:54 UTC+0000  
  
Subkeys:  
  (S) Device Parameters  
  (S) LogConf  
  (S) Properties  
  (V) Control  
  
Values:  
REG_SZ      DeviceDesc      : (S) @disk.inf,%disk_devdesc%;Disk drive  
REG_DWORD    Capabilities    : (S) 0  
REG_MULTI_SZ HardwareID      : (S) ['USBSTOR\DiskGeneral_UDisk_5.00', 'USBSTO  
R\DiskGeneral_UDisk_5.00', 'USBSTOR\DiskGeneral_5.00', 'USBSTOR\General_UDisk_5.00', 'General_UDisk_5.00', 'USBSTOR\GenDisk', 'GenDisk', '', '']  
REG_MULTI_SZ CompatibleIDs  : (S) ['USBSTOR\Disk', 'USBSTOR\RAW', '', '']  
REG_SZ      ContainerID     : (S) {421e7e37-11dd-11e9-b3cf-806e6f6e6963}  
REG_DWORD    ConfigFlags     : (S) 0  
REG_SZ      ClassGUID       : (S) {4d36e967-e325-11ce-bfc1-08002be10318}  
REG_SZ      Class           : (S) DiskDrive  
REG_SZ      Mfg             : (S) @disk.inf,%genmanufacturer%;(Standard disk drives)  
REG_SZ      Service         : (S) disk  
REG_SZ      FriendlyName    : (S) General UDisk USB Device
```

Device categories and class values

Battery Devices

Class = Battery

ClassGuid = {72631e54-78a4-11d0-bcf7-00aa00b7b32a}

This class includes battery devices and UPS devices.

Biometric Device

Class = Biometric

ClassGuid = {53D29EF7-377C-4D14-864B-EB3A85769359}

(Windows Server 2003 and later versions of Windows) This class includes all biometric-based personal identification devices.

Bluetooth Devices

Class = Bluetooth

ClassGuid = {e0cbf06c-cd8b-4647-bb8a-263b43f0f974}

(Windows XP SP1 and later versions of Windows) This class includes all Bluetooth devices.

Camera Device

Class = Camera

ClassGuid = {ca3e7ab9-b4c3-4ae6-8251-579ef933890f}

(Windows 10 version 1709 and later versions of Windows) This class includes universal camera drivers.

CD-ROM Drives

Class = CDROM

ClassGuid = {4d36e965-e325-11ce-bfc1-08002be10318}

This class includes CD-ROM drives, including SCSI CD-ROM drives. By default, the system's CD-ROM class installer also installs a system-supplied CD audio driver and CD-ROM changer driver as Plug and Play filters.

Disk Drives

Class = DiskDrive

ClassGuid = {4d36e967-e325-11ce-bfc1-08002be10318}

This class includes hard disk drives. See also the HDC and SCSIAdapter classes.

Note: It is hardware, difference from Disk interface GUID, which is software

What is ContainerID?

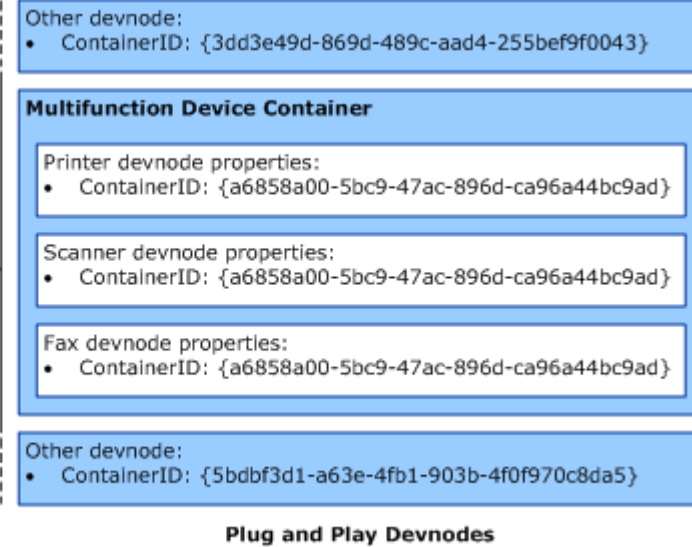
```
Values:
REG_SZ      DeviceDesc      : (S) @disk.inf,%disk_devdesc%;Disk c
REG_DWORD    Capabilities    : (S) 0
REG_MULTI_SZ HardwareID      : (S) ['USBSTOR\\DiskGeneral_UDisk_
'USBSTOR\\DiskGeneral_UDisk_-----', 'USBSTOR\\DiskGeneral_',
ral_UDisk_-----5', 'General_UDisk_-----5', 'USBSTOR\\GenDisk', GenDisk
', ', '']
REG_MULTI_SZ CompatibleIDs   : (S) ['USBSTOR\\Disk', 'USBSTOR\\RAW', '', '']
REG_SZ       ContainerID     : (S) {421e7e37-11dd-11e9-b3cf-806e6f6e6963}
REG_DWORD    ConfigFlags     : (S) 0
```

Windows 7 PC



Multifunction Device

- Printer
- Scanner
- Fax



- A **system-supplied** device identification string
- **Groups** the functional devices associated with a single-function or multifunction device installed in the computer.
- Introduced in Windows 7
 - the Plug and Play (PnP) manager uses the container ID to **group** one or more device
- All physical devices should be assigned a ContainerID – not just storage devices.
 - monitors, external speakers, keyboards

Which drive letter does the USB was assigned to?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 printkey  
-K "MountedDevices"  
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile  
  
-----  
Registry: \REGISTRY\MACHINE\SYSTEM  
Key name: MountedDevices (S)  
Last updated: 2019-01-06 15:02:54 UTC+0000  
REG_BINARY \DosDevices\F: (S)  
0x00000000 5f 00 3f 00 3f 00 5f 00 55 00 53 00 42 00 53 00  
0x00000010 54 00 4f 00 52 00 23 00 44 00 69 00 73 00 6b 00  
0x00000020 26 00 56 00 65 00 6e 00 5f 00 47 00 65 00 6e 00  
0x00000030 65 00 72 00 61 00 6c 00 26 00 50 00 72 00 6f 00  
0x00000040 64 00 5f 00 55 00 44 00 69 00 73 00 6b 00 26 00  
0x00000050 52 00 65 00 76 00 5f 00 35 00 2e 00 30 00 30 00  
0x00000060 23 00 36 00 26 00 31 00 62 00 65 00 63 00 30 00  
0x00000070 66 00 34 00 38 00 26 00 30 00 26 00 5f 00 26 00  
0x00000080 30 00 23 00 7b 00 35 00 33 00 66 00 35 00 36 00  
0x00000090 33 00 30 00 37 00 2d 00 62 00 36 00 62 00 66 00  
0x000000a0 2d 00 31 00 31 00 64 00 30 00 2d 00 39 00 34 00  
0x000000b0 66 00 32 00 2d 00 30 00 30 00 61 00 30 00 63 00  
0x000000c0 39 00 31 00 65 00 66 00 62 00 38 00 62 00 7d 00
```

- ✓ HKEY_LOCAL_MACHINE
 - > BCD00000000
 - > HARDWARE
 - > SAM
 - > SECURITY
 - > SOFTWARE
 - ✓ SYSTEM
 - > ActivationBroker
 - > ControlSet001
 - > CurrentControlSet
 - > DriverDatabase
 - > HardwareConfig
 - > Input
 - > Keyboard Layout
 - > Maps
 - MountedDevices

.?..U.S.B.S.
T.O.R.#.D.i.s.k.
&.V.e.n._.G.e.n.
e.r.a.l.&.P.r.o.
d._.U.D.i.s.k.&
R.e.v._.5...0.0.
#.6.&.1.b.e.c.0.
f.4.8.&.0.&._.&
0.#.{.5.3.f.5.6.
3.0.7.-.b.6.b.f.
-.1.1.d.0.-.9.4.
f.2.-.0.0.a.0.c.
9.1.e.f.b.8.b.}.

SN

Problem of only exam */System/MountedDevices*

- The registry key may not have a correct drive letter.
 - Windows only tracks the last device that has been assigned a particular drive letter (D, E, F, etc.)
- We need to exam NTUSER.DAT
 - Next slide

Who did mount F volume to PC?

Registry Editor

File Edit View Favorites Help

Computer\HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"
```

```
volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile
```

```
-----  
Registry: \??\C:\Users\IEUser\ntuser.dat
```

```
Key name: MountPoints2 (S)
```

```
Last updated: 2019-01-06 15:03:07 UTC+0000
```

```
Subkeys:
```

```
(S) CPC
```

```
(S) F
```

```
(S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
```

```
(S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
```

```
(S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
```

```
(S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
```

```
(S) {a5b8a983-608c-11e5-a266-806e6f6e6963}
```

When the USB Last Attached to PC?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F"  
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile  
  
-----  
Registry: \??\C:\Users\IEUser\ntuser.dat  
Key name: F (S)  
Last updated: 2019-01-06 15:03:06 UTC+0000
```



USB Evidence Source List

- SYSTEM/CurrentControlSet/Enum/USBSTOR
 - Referenced by SYSTEM\ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}\UID\
- SYSTEM/MountedDevices
 - Drive letter or volume mounted
- NTUSER.DAT/Software/Microsoft/Windows/CurrentVersion/Explorer/MountPoints2
 - reveal which user was logged in and active
- SYSTEM/CurrentControlSet/Enum/USB
 - Not available in memory dump
- (optional) Windows Vista+ – ROOT/Windows/inf/setupapi.dev.log
- (optional) Windows XP – ROOT/Windows/setupapi.log



Investigate Internet History

Did the suspect use Internet Explore?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 iehistory | more
Volatility Foundation Volatility Framework 2.6
*****
Process: 2524 explorer.exe
Cache type "URL " at 0x3156080
Record length: 0x200
Location: https://www.google.com/chrome/static/images/favicons/favicon.ico
Last modified: 2018-04-26 05:30:00 UTC+0000
Last accessed: 2018-09-17 14:54:34 UTC+0000
File Offset: 0x200, Data Offset: 0xac, Data Length: 0xbc
File: favicon[2].ico
Data: HTTP/1.1 200 OK
Content-Type: image/x-icon
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Alt-Svc: quic=":443"; ma=2592000; v="44,43,39,35"
Transfer-Encoding: chunked

~U:ieuser

*****
Process: 2524 explorer.exe
Cache type "URL " at 0x3156480
Record length: 0x180
Location: https://www.jam-software.de/css/reset_v4.css
--More--
```

IE history/cache

Control X: exit

Are there any activates in 2019? No

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 iehistory | grep 2019
Volatility Foundation Volatility Framework 2.6
student@kali:~/mem_lab$
```



Investigate File Explorer History

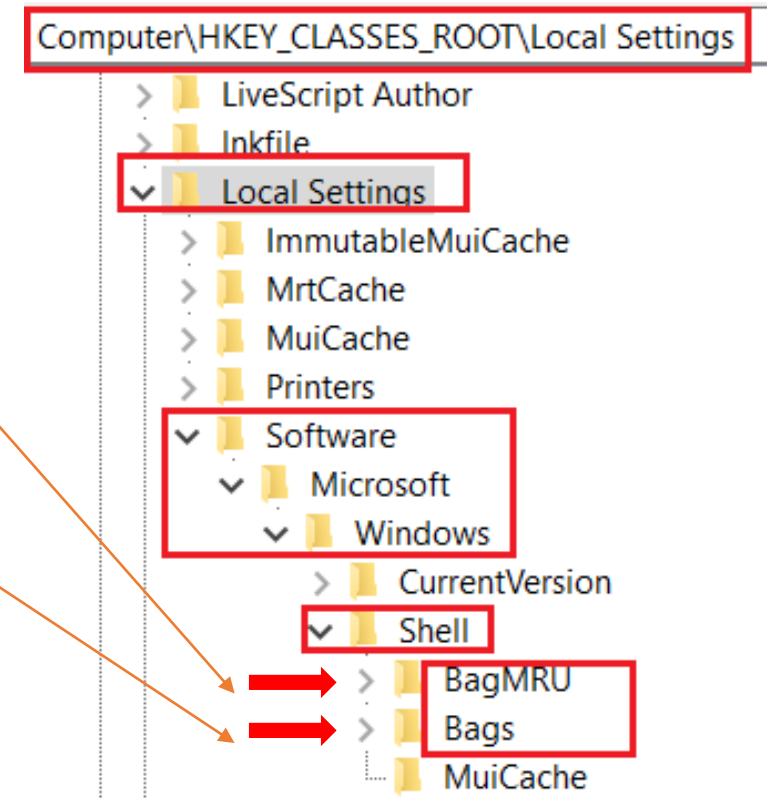
Shellbags

What are Shellbags?

- Registry keys to track a folder window when viewed through File Explorer;
 - views, sizes and positions
 - includes network folders and removable devices
 - available since Windows XP
- Giving the investigator great insight into
 - the folder
 - browsing history of a suspect
 - details for any folder that might no longer exist on a system
 - due to deletion, or being located on a removable device
- Note: shellbags **only** get created when accessing the folders
 - copying, moving, deleted, etc. will not change anything.

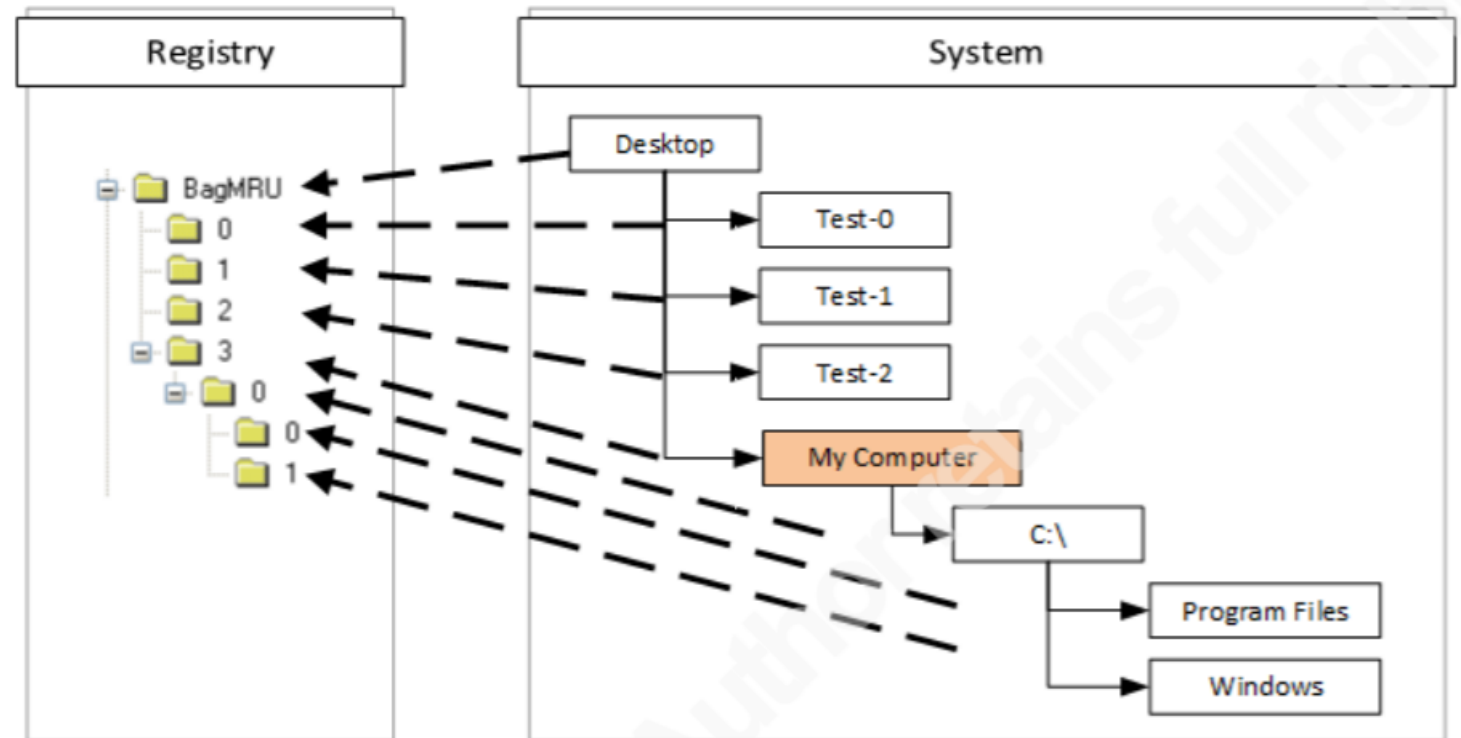
What are registry keys that shellbags are structured from?

- Structured from two main registry keys
 - **BagMRU**: stores folder names and records folder paths by creating the similar tree structure.
 - **Bags**: stores the view preferences such as the window size, location and view mode
- Physical location (Win7 +)
 - %UserProfile%\NTUSER.dat
 - %UserProfile%\AppData\Local\Microsoft\Windows\UsrClass.dat.



How shellbags are structured?

- Structured In hierarchy to which they are accessed through Windows Explorer
 - each numbered folder representing a parent or child folder of the one previous.



Are there any folder access activities on 2019-01-06?

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 shellbags  
| grep "2019-01-06" | sort
```

Volatility Foundation Volatility Framework 2.6

Index	Offset	Operation	Timestamp	Timezone	File Path
1	0	DOWNLO~1	2019-01-06 14:49:40	UTC+0000	2015-09-21 09:17:34 UTC+0000
2019-01-06	14:49:40	UTC+0000	RO, DIR		C:\Users\IEUser\Downloads
Last updated: 2019-01-06 14:32:21 UTC+0000					
Last updated: 2019-01-06 14:32:53 UTC+0000					
Last updated: 2019-01-06 14:55:45 UTC+0000					
Last updated: 2019-01-06 15:04:40 UTC+0000					
Last updated: 2019-01-06 15:04:40 UTC+0000					
Last updated: 2019-01-06 15:04:40 UTC+0000					
Last updated: 2019-01-06 15:07:03 UTC+0000					

Interesting! we need to look into it.

What are folder access activities on 2019-01-06?

Print shellbags to a file shellbags.html

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 shellbags  
--output=html --output-file=shellbags.html
```

```
Volatility Foundation Volatility Framework 2.6  
Scanning for registries....  
Gathering shellbag items and building path tree...  
Outputting to: shellbags.html
```

Easy to review html files

Verify shellbags.html has been generated

```
student@kali:~/mem_lab$ ls shellbags.html  
shellbags.html
```

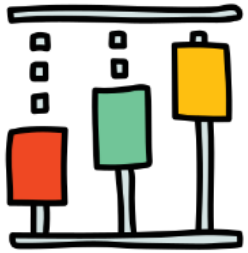
A suspect accesses the download folder (twice).

Key	LastWrite	FileName	Create	Access	Attributes	Unicode	Path
Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Users	2009-07-14 02:37:06 UTC+0000	2015-09-21 09:50:54 UTC+0000	RO, DIR		C:\Users
Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	BGinfo	2015-09-21 09:17:38 UTC+0000	2015-09-21 09:17:38 UTC+0000	DIR		C:\BGinfo
Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Windows	2009-07-14 02:37:06 UTC+0000	2015-09-22 18:34:30 UTC+0000	DIR		C:\Windows
Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Drivers	2015-09-22 08:11:50 UTC+0000	2015-09-22 08:11:50 UTC+0000	DIR		C:\Drivers
Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	PerfLogs	2009-07-14 02:37:06 UTC+0000	2009-07-14 02:37:06 UTC+0000	DIR		C:\PerfLogs
Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0	2019-01-06 15:04:40 UTC+0000	DOWNLO~1	2015-09-21 09:17:34 UTC+0000	2019-01-06 14:49:40 UTC+0000	RO, DIR		C:\Users\IEUser\Downloads

only the longest path is the real folder access. Why?

First Access folder

Last Access of folder or last preference change, not the time of changing the files in the folder



Timeline Analysis

Chrome.exe downloads
2019-01-04 16:05:12 GMT

```
Owner: Process chrome.exe Pid 2388
0x010abb01 31 39 32 2e 31 36 38 2e 35 36 2e 35 2f 73 65 63 192.168.56.5/sec
0x010abb11 72 65 74 5f 66 69 6c 65 2e 64 6f 63 78 22 32 64 ret_file.docx"2d
0x010abb21 30 65 2d 35 37 65 61 34 30 62 34 37 64 39 34 63 0e-57ea40b47d94c
0x010abb31 22 46 72 69 2c 20 30 34 20 4a 61 6e 20 32 30 31 "Fri,.04.Jan.201
0x010abb41 39 20 31 36 3a 30 35 3a 31 32 20 47 4d 54 61 70 9.16:05:12.GMTap
0x010abb51 70 6c 69 63 61 74 69 6f 6e 2f 76 6e 64 2e 6f 70 plication/vnd.op
```

First review download
folder
2019-01-06 14:49:40 UTC

Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	15:04:40 UTC+0000	FEHLogs	02:37:06 UTC+0000	02:37:06 UTC+0000	DIR	CAFEHLogs
<i>Longest path is the real folder access. Why?</i>						
Windows\Shell\BagMRU\0\1\1\0	2019-01-06 15:04:40 UTC+0000	DOWNLO~1	2015-09-21 09:17:34 UTC+0000	2019-01-06 14:49:40 UTC+0000	RO, DIR	C:\Users\IEUser\Downloads

USB plugged in
2019-01-06 15:02:54 UTC

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418
printkey -K "ControlSet001\Enum\USBSTOR"
Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable (V) = Volatile

-----
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000
Subkeys:
(S) Disk&Ven_General&Prod_UDisk&Rev_5.00
```


USB assigned to F volume
2019-01-06 15:03:06 UTC

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418  
printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F"  
Volatility Foundation Volatility Framework 2.6  
Legend: (S) = Stable (V) = Volatile  
  
-----  
Registry: \??\C:\Users\IEUser\ntuser.dat  
Key name: F (S)  
Last updated: 2019-01-06 15:03:06 UTC+0000
```

Last review download folder:
2019-01-06 15:04:40 UTC

Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	15:04:40 UTC+0000	PenLogs	02:37:06 UTC+0000	02:37:06 UTC+0000	DIR	CurrentLogs
<i>only the longest path is the real folder access. Why?</i>						
Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0	2019-01-06 15:04:40 UTC+0000	DOWNLO~1	2015-09-21 09:17:34 UTC+0000	2019-01-06 14:49:40 UTC+0000	RO, DIR	C:\Users\IEUser\Downloads

Command Prompt with Copy command:
2019-01-06 15:06:011 UTC

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 pslist -p 3988  
Volatility Foundation Volatility Framework 2.6  
Offset(V) Name PID PPID Thds Hnds Sess Wow64 Start  
-----  
0x842ddd28 conhost.exe 3988 348 2 52 1 0 2019-01-06 15:06:11 UTC+0000
```

Chrome.exe
downloads:
2019-01-04
16:05:12

USB plugged
in: 2019-01-06
15:02:54 UTC
USB assigned
to F volume:
2019-01-06
15:03:06 UTC

Command
Prompt with
Copy
command
: 2019-01-06
15:06:11 UTC

First review
download folder
: 2019-01-06
14:49:40 UTC

Last review download
folder
: 2019-01-06 15:04:40
UTC

Note

- When a folder is deleted, the correspondent ShellBag registry keys and values will not be removed.
- However, several tools have been developed to remove the ShellBag information from the registry.

Post-Investigation

What is Windows 7 password?

- Dump password hashes from memory
- Using *John the Ripper* to crack password
 - John the Ripper is a free password cracking software tool.
 - *password.lst* is the default password dictionary

Dump NTLM hash

```
student@kali:~/mem_lab$ volatility -f memdumpWin7.mem --profile=Win7SP1x86_23418 hashdump
Volatility Foundation Volatility Framework 2.6
Administrator:500:aad3b435b51404eeaad3b435b51404ee:fc525c9683e8fe067095ba2ddc971889:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
IEUser:1000:aad3b435b51404eeaad3b435b51404ee:fc525c9683e8fe067095ba2ddc971889:::
sshd:1001:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
sshd_server:1002:aad3b435b51404eeaad3b435b51404ee:8d0a16cfc061c3359db455d00ec27035:::
```

The **password** to your VM IEUser is "Passw0rd!"

Save IEUser's password hash to a file

```
student@kali:~/mem_lab$ echo fc525c9683e8fe067095ba2ddc971889 > ntlmhash.txt
student@kali:~/mem_lab$ ls -l
total 1048980
-rw-r--r-- 1 student student 319145 Sep 20 20:03 file.txt
-rw-r--r-- 1 student student 1073676288 Sep 19 17:18 memdumpWin7.mem
-rw-r--r-- 1 student student 33 Sep 22 08:36 ntlmhash.txt
-rw-r--r-- 1 student student 150422 Sep 20 16:52 process_1.txt
student@kali:~/mem_lab$
```


Find the location of John's default password.lst

```
student@kali:~/mem_lab$ sudo find / -type f -name password.lst
/usr/share/metasploit-framework/data/wordlists/password.lst
/usr/share/john/password.lst
find: '/run/user/1001/gvfs': Permission denied
```

/: root f: file

Search passwords contain the string "pass" ignore case

```
student@kali:~/mem_lab$ grep -i pass /usr/share/john/password.lst
#!/comment: This list is based on passwords most commonly seen on a set of Unix
#!/comment: (that is, more common passwords are listed first). It has been
#!/comment: revised to also include common website passwords from public lists
#!/comment: of "top N passwords" from major community website compromises that
password
password1
passion
Password
passw0rd
wordpass
password2
PASSWORD
newpass
passwd
nopass
pass
```

i: ignore case

You can add a new password string “Passw0rd!” to the dictionary

```
student@kali:~/mem_lab$ sudo su
root@kali:/home/student/mem_lab# echo Passw0rd! >> /usr/share/john/password.lst
```

Need root privilege

Verify the password string “Passw0rd!” has been added to the dictionary

```
root@kali:/home/student/mem_lab# exit
exit
student@kali:~/mem_lab$ grep Pass /usr/share/john/password.lst
Password
Passw0rd!
student@kali:~/mem_lab$
```

Find IEUser's password

```
student@kali:~/mem_lab$ john --format=nt ntlmhash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst, rules:Wordlist
Passw0rd! (?)
1g 0:00:00:00 DONE 2/3 (2020-09-22 09:54) 33.33g/s 121600p/s 121600c/s 121600C/s !@#$.sammy
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed
student@kali:~/mem_lab$
```

Try Other Plugins

How to download plugins

Clone a plugin repo

```
student@kali:~/mem_lab$ git clone https://github.com/superponible/volatility-plugins.git
Cloning into 'volatility-plugins'...
remote: Enumerating objects: 91, done.
remote: Total 91 (delta 0), reused 0 (delta 0), pack-reused 91
Unpacking objects: 100% (91/91), 63.47 KiB | 1.59 MiB/s, done.
student@kali:~/mem_lab$ ls
file.txt  memdumpWin7.mem  ntlmhash.txt  process_1.txt  service.html  shellbags.dot  shell
bags.html  volatility-plugins  plugins are stored here
```

Show downloaded plugins

```
student@kali:~/mem_lab$ ls volatility-plugins/
README.md      firefoxhistory.pyc  prefetch.pyc      trustrecords.pyc
apihooksdeep.py  idxparser.py       sqlite_help.py    uninstallinfo.py
apihooksdeep.pyc  idxparser.pyc      sqlite_help.pyc   uninstallinfo.pyc
chromehistory.py  malfinddeep.py     ssdeepscan.py
chromehistory.pyc  malfinddeep.pyc    ssdeepscan.pyc
firefoxhistory.py  prefetch.py         trustrecords.py
```

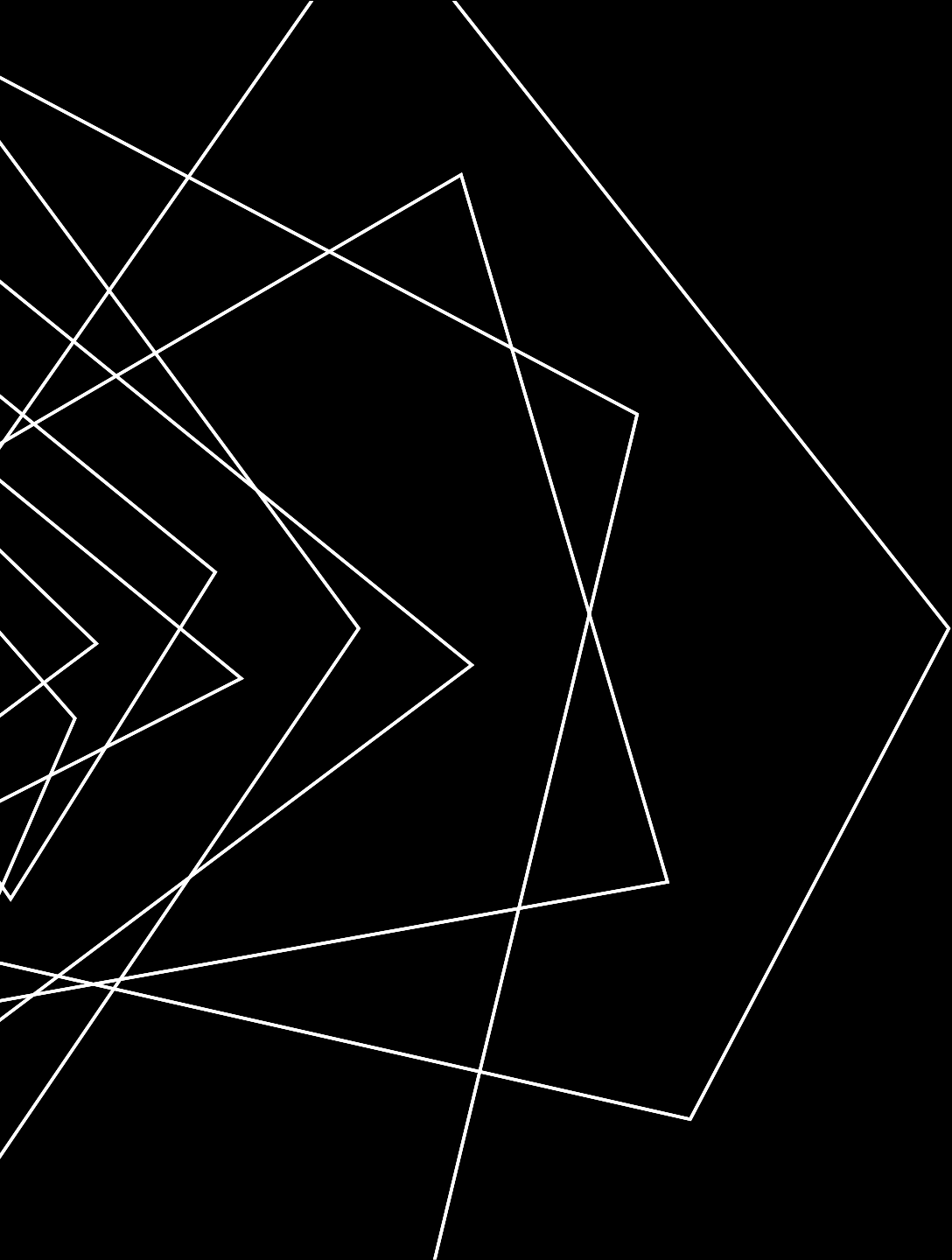
How to find chrome history?

Plugins may not work!

```
student@kali:~/mem_lab$ volatility --plugins=volatility-plugins/ -f memdumpWin7.mem --profile=Win7SP1x86_23418
chromehistory
Volatility Foundation Volatility Framework 2.6
Index  URL
Visits Typed Last Visit Time Title Hidden Favicon ID
-----
```


References

- <https://www.forwarddefense.com/pdfs/Memory-Analysis-with-Volatility.pdf>
- <https://www.andreafortuna.org/2017/07/31/volatility-my-own-cheatsheet-part-6-windows-registry/>
- <http://www.mrtweaks.com/regedit.htm>
- <https://tweaklibrary.com/what-are-registry-root-keys/>
- <https://docs.microsoft.com/en-us/windows-hardware/drivers/usbcon/usb-device-specific-registry-settings>
- <https://docs.microsoft.com/en-us/windows-hardware/drivers/install/standard-usb-identifiers>
- <https://www.forensicfocus.com/forums/general/usbstor-registry-entries-windows-7/>
- <https://docs.microsoft.com/en-us/windows-hardware/drivers/install/instance-ids>
- <http://techexe.net/computer-fundamentals/>
- <https://docs.microsoft.com/en-us/windows-hardware/drivers/install/device-identification-strings>
- https://kuscholarworks.ku.edu/bitstream/handle/1808/19390/Kang_ku_0099M_14318_DATA_1.pdf;jsessionid=8069E32208405DF2EC49129DDE02DF4D?sequence=1
- <http://what-when-how.com/windows-forensic-analysis/registry-analysis-windows-forensic-analysis-part-5/>



Thank you

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